

Highlights

Rv3222c, the unnamed gene between *sigH* and *rshA*, encodes an essential intrinsically disordered protein and is the last intact member of a degenerate gene family in *Mycobacterium tuberculosis*

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- Rv3222c is essential in *M. tuberculosis* but has never been characterized.
- Described unnamed in 1999, it survives in the literature only as a locus-tag error.
- A previously reported extreme CRISPR vulnerability score is an artefact.
- Rv3222c is the sole intact survivor of a gene family split by an IS110 insertion.
- The MTBC state of the family is already present in the closest smooth bacillus.

Rv3222c, the unnamed gene between *sigH* and *rshA*, encodes an essential intrinsically disordered protein and is the last intact member of a degenerate gene family in *Mycobacterium tuberculosis*

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Abstract

Rv3222c is a 183-residue *Mycobacterium tuberculosis* gene with no assigned name and no dedicated study, despite sitting between *rshA* (Rv3221A) and *sigH* (Rv3223c), overlapping both by four nucleotides. A literature search finds dozens of papers on *sigH* and *rshA*, none on Rv3222c, and its only two prior mentions by locus tag are errors. Transposon-sequencing, CRISPR-interference and comparative genomics show Rv3222c is essential in H37Rv; a weaker requirement in East Asian/Indo-Oceanic than in Euro-American strains (Fisher's exact test, $p = 0.036$) coincides with a lineage-specific duplication spanning the locus and cannot yet be distinguished from that artefact. A previously reported extreme CRISPR-interference vulnerability score is an out-of-domain extrapolation from the genome's weakest guides. The protein is 44.8% intrinsically disordered and has no detectable homolog by profile-based methods or structure search against pdb100, CATH50 and the clustered AlphaFold database; its only hit is itself. It is the sole intact survivor of a small gene family: three degenerate copies persist in H37Rv, against at least fourteen in *M. canettii*, tracing to a single ancestral insertion of an IS110-family recombinase into a family member, splitting it into two transposase-flanking halves in every copy. That insertion point falls at the same nucleotide in the 29 of 31 smooth-bacillus and MTBC genomes where it is measurable, placing the event, and the family's contraction to one intact copy, before the MTBC's origin. Co-induction with *sigH* under oxidative stress and an existing chemogenomic hypomorph strain (PROSPECT) position Rv3222c as an experimentally tractable, uncharacterized component of the SigH-RshA locus.

Keywords: *Mycobacterium tuberculosis*, MTBC, Rv3222c, intrinsic disorder, IS110, gene family degeneration, *sigH*, oxidative stress response

1. Introduction

The SigH-RshA module is one of the best-characterized stress-response switches of *Mycobacterium tuberculosis*. SigH is an extracytoplasmic-function sigma factor that reprograms transcription in response to oxidative, nitrosative and heat stress and is required for full virulence; RshA is its cognate anti-sigma factor, sequestering SigH under unstressed conditions and releasing it upon a redox-sensitive conformational change [1, 2]. That switch is not purely redox-driven: the essential kinase PknB phosphorylates both SigH and RshA, and phosphorylation of RshA, but not of SigH, interferes with the interaction between the two in vitro, so that over-expressing *pknB* partially relieves RshA's negative regulation of SigH and raises resistance to oxidative stress [3]. Interest in the module has not abated: SigH activity determines the killing of *Mycobacterium tuberculosis* by nitronaphthofuran prodrugs through induction of the mycoredoxin Mrx2 [4], its ortholog governs intrinsic resistance to several antibiotic classes in *Mycobacterium abscessus* [5], and $\Delta sigH$ strains are the backbone of live attenuated vaccine

candidates now in preclinical evaluation [6]. The two genes are encoded back to back on the minus strand of the chromosome, at Rv3223c (*sigH*) and Rv3221A (*rshA*). Between them, on the same strand, sits a third open reading frame, Rv3222c: 183 residues, overlapping *sigH* and *rshA* by four nucleotides on each side, with no gene name and, in the reference proteome (UniProt O05844), no assigned function beyond "uncharacterized protein".

The contrast within this three-gene arrangement is stark. A title/abstract literature search restricted to the module (76 records) finds 56 papers describing *sigH*, 17 describing *rshA*, and zero describing Rv3222c under any spelling of its locus tag. Rv3222c is also the only one of the three without a gene name, itself evidence that it has never been the object of dedicated study: genes acquire names when someone characterizes them. This asymmetry is not a trivial search artefact. Locus-tag-only searches are known to under-count publications that refer to a gene by name rather than by identifier (*katG*, for instance, returns roughly two orders of magnitude more hits by gene name than by locus tag); but Rv3222c has no gene name to under-count in the first place, so the zero is informative rather than a search limitation.

At the same time, independent lines of genome-wide evidence single out Rv3222c as unusual. Saturating transposon

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mutagenesis calls it essential for growth [7], it is under strong purifying selection within the complex and relative to the out-group *M. canettii*, and a genome-wide CRISPR-interference screen assigns it one of the most extreme vulnerability scores among all uncharacterized *Mycobacterium tuberculosis* genes [8]. It was this last observation, an essential and apparently maximally vulnerable gene with no name and no literature, that originally motivated the present study, conducted as an in silico re-derivation and hardening of an inherited annotation dossier rather than as a new wet-laboratory investigation.

That re-derivation is the subject of this paper, and it did not confirm the dossier as received. Systematically tracing every claim back to its primary source revealed that several of the strongest statements did not survive scrutiny: the CRISPR-interference vulnerability score, on inspection of the underlying per-guide data, is an extrapolation built on guides that are among the weakest in the genome and cannot be cited as evidence of vulnerability; a claim that the gene is restricted to the MTBC turns out to reflect a detection threshold rather than an absence of homologs; and an apparent contradiction between essentiality in the reference strain and transposon-mutant enrichment in clinical isolates turns out to be a labeling artefact of strain-versus-strain comparisons, behind which lies a previously unreported phylogenetic pattern that a genomic duplication confound keeps us from reading as an established lineage-dependent difference (Section 4.4). What survives this process, and what does not, is reported below together with the tests that decided each case. What remains is, we argue, a more interesting and more tightly evidenced story than the one we set out to confirm: Rv3222c is the last coding survivor of a small gene family whose other members were degraded by an ancestral, MTBC-wide fixed insertion of a mobile genetic element, a family history that, together with the gene’s essentiality and its almost complete absence from the literature, motivates its characterization independently of any drug-target framing.

2. Materials and Methods

2.1. Data provenance

Baseline annotation (identity, genomic context, conservation, disorder, structural predictions, functional-partner scores) was obtained from the *annotation_mtbc* atlas (<https://w3id.org/mtbc-atlas/>), a curated, multi-layer per-gene resource built from public MTBC sequencing and genome data; the corresponding record for Rv3222c was resynchronized against its live version immediately before analysis. The strain-level data underlying the conservation, IS6110 and region-of-difference analyses reported below (per-strain variant calls, insertion profiles and locus-level coverage across the genome collections cited at each use) reach the atlas from TB-Annotator [9], a curated compendium of publicly available MTBC sequencing data. Every atlas-derived figure cited in this paper was traced back to its underlying raw count or supplementary data table before use, following the rule that an atlas label is a hypothesis, not a conclusion; two figures inherited from the atlas (a mis-imported quality qualifier on the CRISPR-interference score, and a stale

absence call for non-MTBC mycobacteria) were found on this re-derivation not to hold, and are reported as corrected below rather than as originally stated. The H37Rv reference genome is NC_000962.3 [10]; gene models and legacy functional categories follow Mycobrowser release 5 [11]. Reference genomes used for cross-species comparisons were *M. bovis* AF2122/97 (LT708304.1), *M. africanum* GM041182 (lineage 6), *M. canettii* CIPT 140010059 (NC_015848) and *M. leprae* TN; only complete assemblies were used for family census counts, to avoid assembly fragmentation splitting or duplicating a repeat region.

The bioinformatic analyses, database queries and literature cross-referencing underlying this study were carried out within an AI-augmented bioinformatics working environment, Claude Code (Anthropic), operating the Claude Sonnet 5 language model as an interactive analysis agent under continuous human direction. Every script, statistical test and numerical claim produced within this environment was independently traced back to its underlying raw data, primary publication or database record before being reported here, following the verification discipline described throughout this section; the corresponding author designed the study, specified and reviewed every analysis, and takes full responsibility for the accuracy of its content.

2.2. Essentiality and its mappability

Essentiality follows the saturating HimarI transposon-sequencing analysis of DeJesus et al. [7]. Because Tn-seq calls a gene essential from the absence of observed insertions, a region with non-unique 40-nucleotide windows around its TA dinucleotides could produce a false essential call by multi-mapping rather than by biology. We therefore extracted the coordinates of all TA sites within the Rv3222c coding sequence and tested, for each, whether its flanking 40-nucleotide window is unique in the H37Rv genome (BLAST+ `blastn -task blastn -word_size 7 -dust no`, chosen over the megablast default because the gene’s own paralogous family, described below, diverges enough at the nucleotide level that megablast would miss it) [12].

Non-uniformity of essentiality across the complex was assessed using MtbTnDB, a standardized compendium of Tn-seq conditions [13] that includes eight clinical-strain-versus-H37Rv comparisons from Carey et al. [14]; each comparison’s fold-change sign was reinterpreted in light of its being a strain-versus-strain design (in which a positive log2 fold-change indicates relatively more tolerated insertions, i.e. lower requirement, in the clinical strain, not a treatment-response phenotype), and a two-tailed Fisher’s exact test compared the count of significant versus non-significant comparisons between the Euro-American lineage (which includes H37Rv) and the pooled East Asian/Indo-Oceanic lineages. Rv3222c is itself absent from Carey et al.’s own per-gene resampling output (their supplementary S7 Table), which excludes genes in a region the original authors identified as subject to large-scale, IS6110-mediated duplication in a subset of the eight strains [14]; the MtbTnDB values used here therefore derive from an independent reprocessing of the raw transposon-junction reads, not from the authors’ own gene-level analysis. Cross-referencing the duplication boundaries the original authors report for each strain against Rv3222c’s coordinates

shows that five of the six strains scored significant below carry a duplication spanning the gene, while neither Euro-American strain does; because a duplicated locus is expected to tolerate disproportionately more insertions in a Tn-seq assay independently of any true fitness effect, this is a plausible confound and is discussed as such in Section 4.4.

Independent corroboration that this region is a recurrent site of large-scale duplication across the MTBC, beyond the Carey et al. strains alone, comes from Weiner et al. [15], who mapped duplications spanning Rv3222c in four further clinical strains from an entirely separate dataset: two Beijing/lineage 2 strains carrying a 350-kb duplication (Rv3128c-Rv3427c) and two additional strains carrying overlapping intervals (Rv3193c-Rv3495c and Rv3188-Rv3644), all four encompassing the Rv3221A-Rv3223c operon.

A survey of 1,015 clinical genomes across the three lineages concerned (composition reported with the result below) was used to check whether within-gene or flanking sequence variants could account for the same split. The statistical power of the seven-TA-site essential call was expressed as a binomial probability, $(1 - s)^n$ for n permissive sites at the site-saturation rate s published for permissive sites in the same study [7], evaluated at that source’s optimistic and conservative saturation bounds, rather than by percentile rank alone.

2.3. CRISPR-interference vulnerability

The published Vulnerability Index (VI) and its confidence interval follow Bosch et al. [8]. We additionally retrieved that study’s per-guide dose-response fits (supplementary table mmc2, sheet "Mtb H37Rv two-line fits", 96,700 single-guide RNAs) and its per-gene quality qualifiers (supplementary table mmc3: certain, crispr_ess, n_guides). For every gene we computed the maximum guide “Strength” value, the parameter that determines whether the dose-response model was ever fit in the regime of strong knock-down, and compared Rv3222c’s value to the genome-wide distribution and to the subset of 500 genes the atlas classifies as candidate-druggable ($VI \leq -6$).

2.4. Homology, disorder and structure

Intrinsic disorder was scored with metapredict v3 [16], cross-checked against per-residue AlphaFold and ESMFold pLDDT [17]. Remote homology detection used HHpred against the PDB70/Pfam profile database [18], a local scan of the full Pfam-A.hmm library with pyhmmer at the Pfam gathering (curation) threshold, and Foldseek [19] both locally (PDB and AlphaFold/Swiss-Prot indices) and through the Foldseek web server against pdb100, CATH50, AlphaFold/Swiss-Prot and AFDB50, in 3Di+AA mode, with the ESMFold and AlphaFold (AF-O05844-F1) models used as independent queries; membrane/secretion topology used DeepTMHMM [20]. Absence from the wider genus was tested by tblastn of the Rv3222c protein against 53 assembled non-MTBC *Mycobacterium* genomes, at a relaxed threshold (e-value 10^{-3} , no identity or coverage cutoff, versus the atlas’s original e-value 10^{-5} with $\geq 30\%$ identity and $\geq 50\%$ coverage), with *rshA* and *sigH* run as positive controls in the same search. To rule out a compositional

artefact (Rv3222c is Ala/Gly-rich), each of the three proteins was shuffled 30 times preserving amino-acid composition (seed 20260731) and re-searched under identical parameters; a real, non-compositional signal was defined as a bit score not reached by any shuffle. Partial open reading frames recovered in non-MTBC hits were extended in-frame to the nearest stop codon to assess whether they represent complete coding sequences.

Proteomic detectability was assessed against a whole-proteome null model rather than reported as a bare count. All 3,906 H37Rv coding sequences were translated in silico (NC_000962.3, bacterial translation table 11) and digested in silico with trypsin (cleavage C-terminal to K/R, no missed cleavages); peptides of 7-25 residues, the length range conventionally recovered by bottom-up mass spectrometry, were counted per protein. This count, together with sequence length and K+R content, was compared against detection in each of the 16 individual PaxDb 5.0 datasets available for H37Rv (the aggregate WHOLE_ORGANISM-integrated.txt file was excluded, since it is not an independent observation).

Amino-acid composition of the two disordered regions was read against an organism-matched internal reference rather than against published compositional norms, which are built on eukaryotic proteomes and whose transfer to a bacterial disordered region is not established: all 2,439 metapredict-v3 disordered segments carried by 1,931 of the 3,906 H37Rv genes in the atlas. For each segment we computed the fraction of charged residues, the net charge per residue, the Gly+Ala+Val, aromatic and proline fractions, Kyte-Doolittle hydropathy, Shannon entropy, and the charge-patterning parameter κ of Das and Pappu [21], and expressed the values measured on Rv3222c’s regions as percentiles of the resulting distributions. Per-residue enrichment was tested separately, by a binomial test of each of the twenty amino acids against its whole-proteome frequency, in each of three regions (the two disordered regions and the intervening ordered segment), with Benjamini-Hochberg correction applied across all 60 tests. No fixed low-complexity threshold was imposed, since the reference’s own median entropy (3.30 against a theoretical maximum of 4.32) would place half of H37Rv’s disordered segments below any threshold set above it; complexity is reported by percentile instead. Repeats of three to five residues occurring at least three times were searched across the full sequence.

2.5. Paralogous family and mobile-element architecture

A within-genome paralog search combined sensitive nucleotide (blastn, dust filter disabled) and protein (tblastn) sweeps of the H37Rv genome, with high-scoring pairs merged by positional overlap and annotated against the RefSeq GFF3. The same search, restricted to merging regions within 200 nucleotides and requiring a 50-nucleotide aligned floor per region, was repeated on each complete reference genome to census family copy number. A junction test for the age of the mobile-element insertion searched for genomes in which the two flanking sequences of the insertion site are contiguous (indicating the insertion never occurred, or occurred elsewhere), requiring both flanks to be present on the same contig to avoid mistaking assembly fragmentation for true absence. Element boundaries

were delimited without reference to any existing annotation, by pairwise alignment of copies against one another and identification of the position at which shared homology breaks down. The origin of the repeated unit (single ancestral insertion followed by block transposition, versus independent insertions at a preferred site) was tested by comparing the dispersion of the exact nucleotide cut point across independent copies, together with a control comparing between-copy identity to identity against Rv3222c itself (first over the whole 1,531-nucleotide unit, then, after this first test was found to be confounded by the shared transposase body, repeated on the gene-derived fragments alone). Copies sharing $\geq 400/400$ nucleotides of flanking sequence with another copy were excluded from this test and treated instead as products of a separate segmental duplication. Domain content of the element's internal open reading frames was assessed with the same Pfam/pyhmmer pipeline used for the gene itself.

A paired null model for the purifying-selection signal reported below used the same family. Nucleotide-level mappability of the three degenerate, non-coding regions (occurrence of every 40-nucleotide window on both strands) was established first, as a precondition, before the same collection of per-strain variant calls used for the coding sequence was scanned for the three regions in the same pass, so that distinct-site density per kilobase would be directly comparable across all four regions.

To test whether family content differs within the outgroup rather than only between the outgroup and the complex, the census was extended to every *Mycobacterium canettii* assembly in NCBI at assembly level Chromosome or Complete Genome with a contig N50 of at least 40 kb, a quality floor set before inspecting any result (29 assemblies, one per strain, RefSeq preferred over GenBank). For each genome we recorded the number of related regions, the identity and length of the best block, and the *rshA*-to-*sigH* spacing obtained from the coordinates of the two neighbors' best alignments, so that the state of the locus could be read independently of copy number. Because copy counting is sensitive to assembly fragmentation, the confounder was tested rather than assumed: the rank correlation between contig N50 and region count was computed overall and within each state, and the two states were compared at matched assembly quality. Genome-wide relatedness was measured by ANI with skani. The insertion-junction test above was likewise repeated across this cohort, with the operonic region (delimited by the two neighbors' alignments, extended by 3 kb on each side) analyzed separately from the rest of the genome; cut points are reported in coding-sequence coordinates, Rv3222c lying on the minus strand.

Because that cohort is dominated by one BioProject, its composition was declared and its redundancy tested rather than assumed away. Near-identical strains were grouped by single-linkage clustering on pairwise ANI at three thresholds (99.90, 99.95 and 99.99% identity), one representative per group was retained (highest contig N50), and every conclusion was recomputed on each reduced cohort; the association between the dominant BioProject and the state of the locus was tested by Fisher's exact test before and after reduction. The two states were then placed on a topology: pairwise ANI across the cohort, the two MTBC references and a *Mycobacterium decipiens*

outgroup (skani, -slow) was converted to distance and used to build neighbor-joining and UPGMA trees, on which the binary character was scored under Fitch parsimony, whose score is invariant to rooting, and under an irreversible model counting acquisitions of the derived state only. Two null models frame that count: 10,000 random permutations of the states across the tips, giving the distribution expected in the absence of phylogenetic structure, and 200 parametric bootstrap replicates in which each pairwise ANI is redrawn from its skani confidence interval and the tree rebuilt, giving the sensitivity of the count to the noise of the distance estimate. The outgroup, which carries no member of the family, was scored as an uninformative state so that it neither forces nor prevents a transition.

2.6. Transcriptional architecture of the locus

Whether Rv3222c can be transcribed independently of *sigH* was assessed against published transcription start site maps rather than by prediction. The supplementary tables of Cortes et al. [22], covering exponential growth and growth arrest, and of Shell et al. [23] were retrieved through Europe PMC and searched for every mapped position falling in a 5-kb window around the operon, retaining the strand only from the tables that list raw positions, since the per-gene tables report the strand of the gene rather than of the start site. The positions found were then classified against the coordinates of the three genes and against the portions of *sigH* that the published $\Delta sigH$ construction leaves in place [24]. The mapping of *sigH*'s own primary start site, recovered by both studies at the same position, serves as the positive control validating the coordinate frame.

A direct test for polar essentiality compared, at each of the three genes, the fraction of TA sites with recovered insertions and the mean non-zero read count from the same saturating Tn-seq dataset [7], re-queried live rather than taken from any intermediate summary. The start and stop codons flanking Rv3222c on each side were checked for overlap against the RefSeq GFF3 coordinates of every co-oriented, adjacent CDS pair genome-wide, using the full CDS span rather than the shorter codon-level description, to build a null distribution for the frequency of the observed 4-nucleotide overlap, evaluated once per junction and once for its joint occurrence at both flanks of the same gene among the subset of H37Rv genes with a co-oriented neighbor on each side. Rho-dependent and Rho-independent transcription termination sites around the locus were retrieved from the supplementary tables of D'Halluin et al. [25], positioned in the coordinate frame validated in the previous paragraph, and compared against the fraction of length-matched H37Rv coding sequences (450-650 nt) with no internal termination site reported in the same dataset, to calibrate whether an absence at this locus is locus-specific or typical for a coding sequence of this size.

2.7. Regulatory context and literature

Regulatory relationships and functional-partner scores follow the Institute for Systems Biology's signed *Mycobacterium tuberculosis* transcriptional regulatory network [26, 27] and STRING, respectively, with STRING's channel-resolved scores (experimental, database, textmining, coexpression,

neighborhood) inspected individually rather than relying on the combined score alone. Because that channel is an aggregate, curated interaction data were also queried at the source: IntAct was searched by UniProt accession for Rv3222c (O05844) and, as positive controls, for SigH (P9WGH9) and RshA (P9WJ69). BioGRID was not queried directly, its API requiring a registered access key; it is covered indirectly, since STRING’s experimental channel aggregates it. Operon prediction follows the co-directional intergenic-distance criterion of Salgado et al. [28]. Literature coverage was assessed with a title/abstract sweep of PubMed under every known locus-tag alias (H37Rv, *M. bovis*, *M. marinum*, *M. smegmatis*, *M. leprae*, *M. abscessus*), followed by a full-text search (Europe PMC) of individual candidate articles and of the supplementary data of Bond et al. [29], since locus tags frequently occur in article bodies and data tables without appearing in the title or abstract. Candidate literature errors were confirmed by reference chaining (tracing a citation to its stated source) and by an independent check unrelated to any locus-tag annotation, namely the stated length of the protein under discussion.

3. Results

3.1. Rv3222c is a genomic blind spot between two well-studied, named neighbors

Rv3222c lies on the minus strand of the chromosome between Rv3221A (*rshA*) and Rv3223c (*sigH*), overlapping each by four nucleotides, an arrangement consistent with, though not proof of, co-transcription. The Institute for Systems Biology’s signed regulatory network places Rv3222c under the control of a single transcription factor, SigH itself [26], and its highest-scoring STRING functional partner is RshA (combined score 979) followed immediately by SigH (967); both scores, however, are driven almost entirely by the neighborhood and coexpression channels, which encode the same genomic-adjacency fact as the operon structure itself, and neither partner shows any experimental or curated-database evidence. Querying a curated interaction database directly, rather than relying on STRING’s aggregation of one, returns the same answer: IntAct holds no interaction record of any kind for Rv3222c (O05844). The same query run on its two neighbors, as a control that the database is not simply empty at this locus, returns two records each, both from a single study [3]: the SigH-RshA complex itself, by comigration on a non-denaturing gel, and each of the two proteins with the phosphatase PstP. Two independent routes, STRING’s experimental channel and IntAct queried directly, thus agree that no physical interaction has ever been recorded for Rv3222c, which for a protein no publication has ever addressed is an absence of evidence rather than evidence of absence.

Genomic co-localization and STRING context scores therefore describe one fact, not two independent lines of evidence.

Despite this position at the heart of one of the best-studied stress-response loci of *Mycobacterium tuberculosis*, Rv3222c has never itself been the subject of a publication. To test that the title/abstract asymmetry reported in the Introduction is not an

artifact of indexing, we assembled the module’s literature at full text: 335 Europe PMC records matching five complementary queries on *sigH*, *rshA*, mycobacterial anti-sigma factors and the locus tags of the three genes, of which 223 full texts could be retrieved and passed a positive control requiring the string *sigH* or *RshA* to be present, an unretrieved or empty document being scored unusable rather than counted as an absence. Across those 223 papers, the string Rv3222 appears exactly twice, and neither occurrence is a study of the gene: one is the transcriptomic table discussed below, the other is a misspelled locus tag denoting its neighbor. Full-text mining across the whole of Europe PMC, not restricted to the module, likewise recovers two articles carrying Rv3222c itself, and both trace to the same source.

The gene was nonetheless described, once, at the very beginning. The paper that discovered *sigH* in mycobacteria [30] maps the locus and reports that, in *Mycobacterium tuberculosis*, “an open reading frame of 546 bases is similarly oriented and has an initiation codon overlapping the *sigH* stop codon”, that “the deduced amino acid sequence is highly similar to that of another *M. tuberculosis* protein of unknown function”, and that “it is not similar to the *M. smegmatis* sequence in the corresponding location”. No other gene in the genome answers that description. Three observations of the present work are therefore already latent in that 1999 paragraph: the intercalation itself, the existence of a paralogous relative of unknown function (Section 3.6), and the absence of the intercalated frame outside the complex (Section 3.6). The gene was seen, correctly placed, and left unnamed. Three years later it travelled further still without being noticed: the complementation plasmid used to restore the *sigH* deletion in CDC1551 carried “the complete coding sequences of three distal genes (MT3319, MT3318, and MT3317)” [24], and MT3319 is Rv3222c, its 183-residue product identical to the H37Rv sequence. Our gene has been cloned into a published vector, transformed into *Mycobacterium tuberculosis*, and reported in *PNAS*, without being named or discussed once.

What the module’s literature does state, consistently, is that the operon has two genes. Of the 223 full texts, 35 describe the arrangement, and they describe it identically over twenty-five years and across four bacterial genera: “the *sigH* gene forms an operon together with the gene *rshA*”; “*Mtb sigH* exists in an operon with an antisigma factor *rshA*”; “the gene encoding RshA lies in the same operon as SigH”. In *Corynebacterium glutamicum*, where the operon has been characterized transcriptionally in detail [31], and in *Streptomyces coelicolor*, where the homologous pair is *sigR-rsrA*, the operon is indeed bicistronic. In *Mycobacterium tuberculosis* it is not: a third reading frame of 552 nt sits between the two, overlapping each by four base pairs, and it appears in none of these descriptions. Thakur et al. [32] list, among structurally characterized sigma/anti-sigma pairs, “ σ^H -Rv3222c”; but the anti-sigma factor of SigH is RshA, encoded at Rv3221A, not Rv3222c, and the reference Thakur et al. cite for this pairing is Song et al. [1], the paper establishing RshA as SigH’s anti-sigma factor. Independently of any locus-tag annotation, Song et al.’s own abstract [1] states that RshA is a 101-residue protein; Rv3222c is 183 residues. The mismatch is unambiguous and the other four sigma/anti-sigma pairs listed in the same table by Thakur et al. are correct, in-

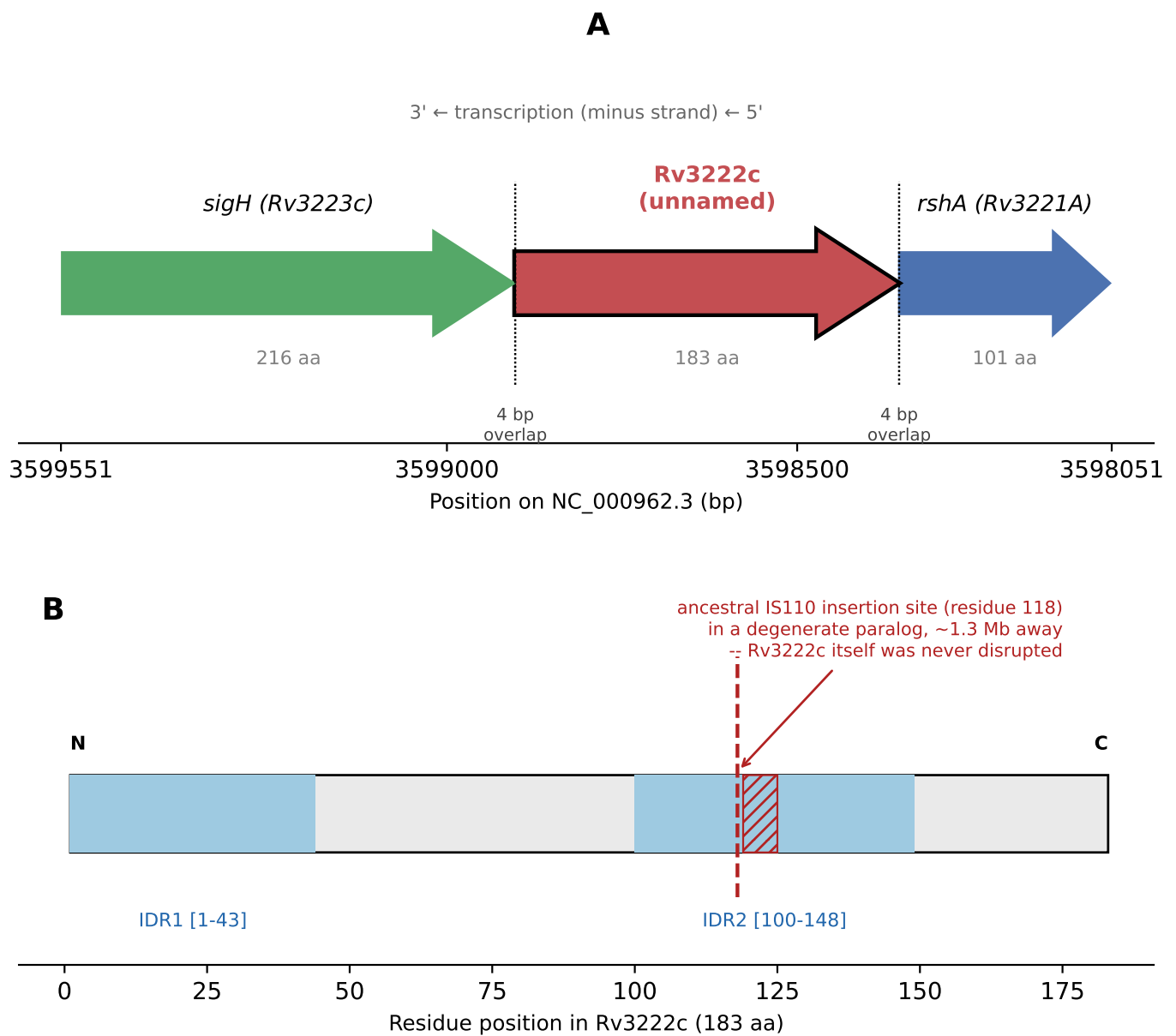


Figure 1: **Rv3222c** between *rshA* and *sigH*. (A) Genomic arrangement at the H37Rv locus (NC_000962.3), all three genes on the minus strand, with the two 4-nucleotide overlaps that flank Rv3222c. (B) Schematic of the Rv3222c polypeptide (183 residues) with its two metapredict-v3 disordered regions and the residue position (118) at which an IS110-family element inserted into a paralogous copy of this gene family elsewhere in the genome (~1.3 Mb away, Section 3.6); Rv3222c itself, at this locus, was never disrupted by that event.

dicating an isolated transcription error between two adjacent locus tags rather than a systematic problem. The second mention, in a transcriptomic study of cigarette-smoke-condensate exposure, annotates Rv3222c as “sigma factor associated” while citing Thakur et al. as its source [33]: an inherited error, though the differential-expression measurement in the same study is independent of it and is discussed separately below.

Locus-tag confusion at this position is in fact older than that 2007 table. The 2002 *sigH* mutant paper reports the expression of “gene MT3318 (Rv3222Ac), which is located distal to *sigH* in the same transcriptional unit and exhibits homology to the anti- σ factor gene class” [24]. MT3318 is RshA, a 101-residue protein whose locus tag is Rv3221A; the tag “Rv3222Ac” exists in no annotation, past or present, and combines its neighbor’s number with two suffixes. The measurement itself, a 4.2-fold reduction of *rshA* in the *sigH* mutant, is unaffected; what the episode shows is that the unnamed gene’s number has been drifting onto its named neighbor for more than twenty years.

This locus-tag substitution has propagated further than two citations. Its clearest and most recent trace is in the supplementary data of a 2025 chemogenomic screening platform, whose data identifier for this locus pairs the Rv3222c locus tag with the *rshA* gene name in a single field [29]. At the same time, its reach is bounded in a specific and informative way. The UniProt record for Rv3222c (O05844) carries no trace of the error, and of the 23 articles that cite Thakur et al. [32], including the field’s major reviews on extracytoplasmic-function sigma factors, none reproduces the erroneous locus tag; they refer to “RshA” by name. The error propagates specifically where locus tags are manipulated computationally, a transcriptomic annotation table, a chemogenomic data identifier, and not in prose review literature, where an anti-sigma factor is referred to by its established name. Rv3222c is precisely the one gene in its own operon without a name to protect it from this kind of transcription error, which is itself a symptom of the same underlying blind spot: a gene studied only through its neighbors, cited only by a typographical accident, and never characterized on its own terms.

3.2. Essentiality is genuine, but not uniform across the complex

Saturating transposon mutagenesis of H37Rv calls Rv3222c essential: no insertion is recovered at any of its seven TA dinucleotide sites [7]. Because an essential call rests on the absence of reads, and reads that map to more than one genomic location are routinely discarded before analysis, an artefactually essential call can arise from multi-mapping alone if a gene’s sequence is locally repeated elsewhere in the genome, which, as described in Section 3.6, Rv3222c’s is. We tested this directly: all seven TA sites of Rv3222c sit inside a 40-nucleotide window that is unique across the H37Rv genome (0/7 ambiguous). The nucleotide-level divergence between Rv3222c and its degenerate paralogs (74 to 90% identity, see Section 3.6) is large enough that no 40-nucleotide read from a true Rv3222c insertion could be confused with one from a paralog. The essential call is therefore a genuine biological signal, not a mappability artefact.

Two qualifications are worth stating rather than omitting. First, seven TA sites is on the low side: the genome-wide median is 13 per gene (19th percentile; 24% of the 3,978 annotated

genes have as few or fewer), which limits statistical power without being exceptional, and is comparable to the gene’s immediate neighbors (*rshA*, 3 sites; *sigH*, 15 sites), neither of which is treated with reservation in the literature. That comparison by rank, however, understates the strength of the call once expressed as a probability. Six of the seven sites are classified permissive at the coverage used; taking the site-saturation rates for permissive sites published in the same study [7], the probability of recovering zero insertions across all six under the null hypothesis that the gene is not essential is 3.0×10^{-12} at the optimistic bound (98.8% saturation, permissive sites in high-coverage non-essential regions) and 1.8×10^{-7} at the conservative bound (92.5%, all sites classified as putatively non-essential pooled). The two bounds differ by a factor of roughly 60,000, yet both remain many orders of magnitude below any conventional threshold, including a Bonferroni correction across the genome’s approximately 4,000 annotated open reading frames ($\sim 1.25 \times 10^{-5}$); the calculation assumes independence between TA sites within one small ORF, a standard simplification in this literature rather than an exact model, but the qualitative conclusion does not depend on it given the margin involved. Second, of the 533 twenty-nucleotide CRISPR-interference protospacers that could be designed against Rv3222c, 22 (4.1%) have an exact second target site elsewhere in the genome, clustered in four blocks (residues 67-75, 125-132, 133-142, 175-185); this does not invalidate the essential call itself, but any single-guide result would need to be checked against the specific guides used before being cited as clean.

Essentiality, however, is not uniform across the MTBC. Six of eight strain-versus-H37Rv transposon comparisons in the MtbTnDB compendium are statistically significant for Rv3222c, but their sign requires care: these are strain-versus-strain designs [14], not treatment-versus-control ones, so a positive log₂ fold-change signifies more insertions tolerated in the clinical strain than in H37Rv, i.e. a lower requirement for the gene, not (as the automated label originally applied to these data stated) a fitness advantage from disruption; the two readings are compatible, and once corrected, the clinical-strain data reinforce rather than contradict H37Rv essentiality, since a large ratio of tolerated-to-baseline insertions still implies very few insertions at baseline.

What the corrected data reveal instead is a phylogenetic pattern: the six significant comparisons (log₂FC 5.40-7.49, $q \leq 0.014$) are the four East Asian and two Indo-Oceanic strains tested, while the two Euro-American strains, the lineage to which H37Rv itself belongs, are not significant (log₂FC 1.81 and 1.98, $q \approx 0.4$), with no overlap between the two groups (Fisher’s exact test, $p = 0.036$). Rv3222c is essential in H37Rv and, by extension, in the lineage 4 background it represents, and significantly less required in lineage 1 and lineage 2 backgrounds. This pattern carries a caveat detailed in Methods and revisited in Section 4.4: Rv3222c falls in a genomic region Carey et al. themselves excluded from their own per-gene analysis as subject to large-scale duplication in a subset of their clinical strains, and that duplication spans Rv3222c in five of the six strains scored significant here but in neither Euro-American strain, so a copy-number artefact cannot be excluded as a full or partial driver of the result.

A survey of 1,015 genomes across the three lineages (180 lineage 1, 82 lineage 2, 753 lineage 4) found only seven distinct variants at the locus and its flanks, none within the coding sequence at appreciable frequency and the single common one (26% of lineage 1) both non-coding and absent from lineage 2, which shows the same reduced requirement; the protein itself is essentially invariant across the three lineages (1 non-synonymous against 5 synonymous variants genome-wide, $\pi_N/\pi_S = 0.074$, 95% exact-binomial CI 0.00-0.51 on this small count, one-sided test against neutrality $p = 0.0067$), so the differential requirement, if genuine rather than a duplication artefact (Section 4.4), is not explained by variation at the locus and must instead reflect the genetic background elsewhere in the genome. This scarcity of non-synonymous change is not an artefact of a poorly covered or slowly mutating locus: the gene's total variant density, synonymous and non-synonymous combined, sits at the 76th percentile of the H37Rv proteome, so the locus is mutated, and evidently sequenced, at a normal rate; it is specifically missense variation that is suppressed.

A second, paired control situates that density against the gene's own decayed relatives rather than against the proteome average. Scanned in the same pass across a 142,984-genome collection, and after confirming that all four regions are individually unique at 40-nucleotide resolution, which rules out a mismatching explanation for any difference observed, Rv3222c carries fewer distinct polymorphic sites per kilobase (682.97/kb) than any of the three degenerate, non-coding copies of its own family described in Section 3.6: 4.4% fewer than the intergenic vestige (714.29/kb), 11.3% fewer than the IS1607 C-terminal fragment (769.91/kb), and 29.5% fewer than the IS1607 N-terminal fragment (968.93/kb), the most divergent and presumably most anciently decayed of the three. Rv3222c is therefore conserved not only relative to the genome-wide background above, but relative to its own family's decayed, non-coding relics in the same genome, the paired null model a purifying-selection claim of this kind requires.

A third, orthogonal control situates the same conservation claim in a different class of mutation entirely: large structural variants rather than point substitutions. Across 5,211 modern strains surveyed for IS6110 insertions, RD-type deletions and locus-level coverage dropouts, Rv3222c registers zero disruptive event of any of the three classes (0/5,211 strains flagged as missing the gene; binomial $p \approx 4 \times 10^{-34}$ against the genome-wide background rate of 1.465% per gene). The intergenic vestige matches this pattern exactly (0 IS6110 insertions), while the two IS1607-derived vestiges carry a small number of raw IS6110 insertions (3 and 1, respectively) that remain far below the count expected from regional IS6110 density once local context is taken into account (16.0 and 16.7 expected; local $p \approx 0$ for both), i.e. markedly depleted rather than strictly absent. Resolving the mechanism behind the phylogenetic pattern, and ruling out the duplication confound that might instead explain it, would require a genetic-interaction screen and a duplication-aware reanalysis, both outside the scope of this study; we report the pattern here as an intriguing, unresolved finding rather than a settled one (Section 4.4).

A separate line of evidence, independent of both Tn-seq

platforms above, corroborates essentiality: the PROSPECT chemogenomic platform has already constructed and screened a Rv3222c depletion strain (Rv3222c-FLAG/DAS+pTetON-10 sspB) as part of a panel in which every strain is, by design, depleted of a different essential protein [29]. The existence of this strain, built and maintained by a group independent of the Tn-seq studies above, is itself a corroborating data point; its further chemogenomic properties are discussed in Section 4.2.

3.3. Viable $\Delta sigH$ mutants constrain how the locus must be transcribed

An essential gene immediately downstream of a dispensable one raises an obvious objection. On the minus strand the order of transcription is *sigH*, then Rv3222c, then *rshA*, so Rv3222c lies downstream of *sigH* within the predicted operon; yet *sigH* mutants are viable, grow indistinguishably from wild type in vitro, and now serve as the backbone of live attenuated vaccine candidates in preclinical evaluation [24, 6]. If Rv3222c is essential, how do these strains survive?

Published transcription start site maps answer the first half of the question, and they answer it negatively. Across two independent studies and three conditions, no transcription start site is detected on the minus strand anywhere between the initiation codon of Rv3222c (3,598,904) and that of *sigH* (3,599,551), nor between *rshA* and Rv3222c. The locus is transcribed from a single dominant start site 127 to 128 nucleotides upstream of *sigH*, recovered in exponential growth and in growth arrest by one study and independently by the other, with two weaker upstream sites also reported [22, 23]. The one start site mapped inside that interval, at 3,599,338, lies on the plus strand and is classified as antisense to *sigH* in both conditions, so it cannot generate an Rv3222c transcript. Rv3222c therefore has no promoter of its own in the conditions mapped so far, and depends on read-through from the *sigH* promoter. The qualification matters: these maps cover exponential growth and nutrient starvation, not oxidative or thermal stress, which is precisely the regime in which SigH is active. Absence of a detected start site is not absence of a promoter.

The geometry of the mutants supplies the second half. The founding $\Delta sigH$ strain is not a clean deletion of the gene: allelic exchange removes the central portion of *sigH* while "leaving 82 bp of the 5' end and 113 bp of the 3' end of the *sigH* gene in place" and inserting a hygromycin cassette between them [24]. On the minus strand it is the 3' end of *sigH* that abuts Rv3222c, so the native promoter and its start site are untouched by the construction, but transcription must now traverse the cassette to reach the essential gene. Given a viable mutant with no internal promoter mapped, at least one of three statements must hold, and each is informative: transcription reads through the cassette and Rv3222c remains expressed; an internal promoter exists but is active only under conditions the available maps do not cover; or the requirement for Rv3222c depends on genetic background. The third is the weakest of the three, because a viable *sigH* null mutant also exists in the H37Rv background (strain ST49), the very background in which the transposon essentiality call was made [34, 4], and because a third, independent construction

replacing *sigH* with an apramycin cassette by phage transduction likewise yields strains with normal in vitro growth [6].

An orthogonal test of the read-through hypothesis is available in the same saturating Tn-seq dataset that calls Rv3222c essential, independent of any single mutant's construction. *sigH* itself, immediately upstream, is called non-essential, with insertions recovered at all 15 of its TA sites (mean non-zero read count 127.9); *rshA*, immediately downstream of Rv3222c, is not called non-essential like *sigH* but Uncertain, with insertions recovered at only 1 of its 3 TA sites, 2 of the 3 in an essential state; an independent method, CRISPR-interference, gives *rshA* a confident essential call (Vulnerability Index -9.327, 95% CI -11.012 to -7.469, certain = True, 20 guides) [8], in contrast to the low-confidence extrapolation retracted for Rv3222c itself in Section 3.4. If any transposon insertion silenced everything transcribed downstream of it, insertions across *sigH* would be lethal by polarity on the essential gene immediately following it, and on this evidence they are not; conversely, a confidently essential anti-sigma factor immediately downstream of Rv3222c is exactly the configuration in which a polar effect, had one existed, would have shown up. DeJesus et al. apply the identical reasoning within the same study to a structurally analogous case, the small RNA ncRv12783c, which shares a transcription start site with the essential gene Rv2783c immediately downstream: a tolerated TA site between the two is read there as supporting "the possibility that Rv2783c and ncRv12783c are independently essential RNA species", not as evidence against Rv2783c [7].

One caveat keeps the mechanism, though not the empirical result, open. The magellan4/Himar1 mariner system underlying this dataset, and the resistance cassettes used to construct the $\Delta sigH$ mutants above, can itself drive outward transcription of a downstream gene from the cassette's own promoter when inserted in the matching orientation, a property the same system's founding paper demonstrates directly in a mycobacterium: a magellan4 insertion upstream of, and co-oriented with, the phage-resistance gene *mpr* in *Mycobacterium smegmatis* conferred a resistance phenotype attributed to the kanamycin cassette's own promoter driving *mpr* overexpression, a reading the authors relate to a similarly oriented, then-unpublished effect in *Haemophilus influenzae* operons [35]; the published Tn-seq data used here do not resolve insertions by orientation, so whether tolerance of insertions in *sigH* reflects native read-through of the locus or the inserted element's own promoter cannot be settled from these data alone. What the test does establish, independently of that open mechanistic question, is that an insertion upstream of Rv3222c and *rshA* does not by itself abolish viability, which rules out the simplest form of the polar-effect objection, that any transposon insertion silences everything downstream of it.

The locus also carries a structural signature of tight coupling that had not previously been quantified against a genome-wide baseline. Both of Rv3222c's flanks show a 4-nucleotide overlap between adjacent coding sequences (Figure 1), the canonical start/stop-codon overlap (ATGA at the *sigH*/Rv3222c junction; GTGA, with a weak GTG start, at the Rv3222c/*rshA* junction) associated with translationally coupled gene pairs. Measured against every co-oriented, adjacent CDS pair in H37Rv (2,567

pairs, no other gene interposed), this exact 4-nucleotide overlap is a minority but not a rare event at a single junction, shared by 503 of them (19.6%); what is unusual is that Rv3222c carries it at both flanks simultaneously, a configuration found in only 102 of the 1,761 H37Rv genes with a co-oriented neighbor on each side (5.79%). An overlap is not itself evidence of coupling, which only a reporter-fusion assay could establish, and we draw no mechanistic conclusion from it beyond what the two figures state: an isolated junction of this kind is unremarkable, but the same junction recurring on both sides of one small gene is not the genomic default.

A third, independent line of transcriptional evidence is negative but uninformative on its own without a calibration the raw absence does not provide. Genome-wide Term-seq mapping of premature Rho-dependent and Rho-independent transcription termination in *Mycobacterium tuberculosis* [25] reports none, of either class, on the minus strand within Rv3222c or in the intervals between Rv3222c and *rshA* or between Rv3222c and *sigH*, consistent with uninterrupted transcription through the operon. Taken alone this would be an appealing corroboration of read-through, but 409 of 534 H37Rv coding sequences of comparable length (450-650 nt) likewise report no internal termination site in the same dataset (76.6%), so the absence at this locus is the typical outcome for a short gene rather than a locus-specific signal, and the Term-seq data available were collected in standard growth, not the oxidative-stress condition under which SigH, and by extension this operon, is active. We report this as a negative result consistent with, but not corroborating beyond its own limited sensitivity, the transcriptional architecture described above.

A fourth, more direct line of evidence comes from strand-specific SEnd-seq and NET-SEnd-seq mapping of RNA 5' and 3' ends genome-wide [36] (GEO GSE211992), which recovers the operon's transcription start site at single-nucleotide resolution in 10 of 13 samples and directly measures how much of the initiated RNA polymerase reaches each downstream gene. Coverage falls to 22-31% of its start-site value by the time transcription reaches the Rv3222c initiation codon (774 nt downstream) and to 4-9% at the Rv3222c/*rshA* junction (1,325 nt downstream); measured against the empirical distribution of survival at the same two distances across 2,369-2,853 (774 nt) and 2,368-2,852 (1,325 nt) other H37Rv transcription start sites in the same samples, however, the locus sits at the 58th-65th and 38th-44th percentiles respectively, an ordinary rather than an exceptional level of attenuation, and no discrete pause site is detectable within the operon (the largest 3'-end peak carries 1.7% of the local signal mass). This dataset does not corroborate the translational-coupling mechanism discussed above beyond what is already stated: RNA polymerase density shows no local accumulation at either the *sigH*/Rv3222c or Rv3222c/*rshA* junction (0.6% of local mass at the one 3' peak inside Rv3222c), so read-through here appears to reflect the locus's ordinary transcriptional attrition rather than a junction-specific pause resolved by coupled translation. The measurement is nonetheless the most direct evidence available that a substantial fraction, roughly a quarter, of RNA polymerases initiating at the operon promoter never reach Rv3222c, and fewer than one in ten reach *rshA*, which sharpens

rather than answers the question of what sustains sufficient expression of an essential gene under this promoter architecture. As with the Term-seq data above, these measurements come from standard growth conditions, not the oxidative-stress regime in which SigH is active.

The complementation constructs are informative in the same way, and the 2002 plasmid discussed in Section 3.1 is not an isolated case. The second group to complement a *sigH* mutant used a 2,022-bp fragment described as carrying *sigH* and its anti-sigma factor *rshA* [4]; since *rshA* and *sigH* physically flank Rv3222c, no contiguous fragment carrying both can exclude the gene between them. Two independent laboratories therefore restored the whole operon rather than *sigH* alone, which is what a polar effect on an essential downstream gene would require, and neither named the gene they were carrying along. Whether or not the inclusion was deliberate, the prediction it implies, that Rv3222c must remain expressed in every viable $\Delta sigH$ strain, is directly testable by RT-qPCR on strains that already exist.

3.4. Rv3222c is not, however, established as CRISPR-interference vulnerable

A genome-wide CRISPR-interference screen reports a Vulnerability Index of -15.542 (95% CI -17.972 to -12.956) for Rv3222c, among the most extreme values recorded for any uncharacterized gene in the dataset [8], and this score was the original motivation for the present study. On inspection of the underlying per-guide dose-response data, this score does not support the claim it was used to make. The source table itself flags the estimate as uncertain (`certain = False`) and reports that the same study's own CRISPR-interference-based essentiality classifier does not call the gene essential (`crispr_ess = NonEssential`), a direct tension with an index at the extreme end of the vulnerability distribution; only 14 guides were scored for this gene, against a median of 90 among genes flagged `certain = True` (*rpoB*, for comparison, has 308). The uncertainty flag alone is not discriminating (it applies to 85% of all genes), but the two underlying populations differ sharply: median VI is -6.76 among `certain = True` genes and $+0.69$, centered on no signal, among `certain = False` genes, and extreme vulnerability scores are markedly enriched among the certain population.

The explanation is direct rather than merely statistical. Among the 96,700 single-guide dose-response fits underlying the screen, Rv3222c's strongest guide reaches a knock-down "Strength" of 0.430, against a genome-wide median, per gene, of 0.960 for the best guide; only 13 of 4,054 scored genes have a weaker best guide (0th percentile). The model linking knock-down strength to fitness cost was therefore never fit in the regime of strong depletion for this gene, and its extreme VI is an extrapolation outside the range in which it was trained, not a measurement. The same defect affects a non-trivial fraction of the atlas's broader "candidate-druggable" gene set ($VI \leq -6$, $n = 500$): 7.4% of it has a best-guide strength below 0.6, and Rv3222c's own value of 0.430 is lower than all but 9 of those 500 genes. The essential-versus-vulnerable distinction matters here: nothing in this section affects the Tn-seq essentiality call of Section 3.2, which rests on different data and has separately

withstood the mappability test above. What it removes is the claim, present in earlier framings of this project, that Rv3222c is among the most vulnerable genes in the genome; that claim is retracted, and Rv3222c is reported here as essential, not as vulnerable.

3.5. A short, intrinsically disordered protein with no detectable homolog

Rv3222c (18.3 kDa, isoelectric point 6.06) is predicted 44.8% intrinsically disordered by consensus metapredict/pLDDT scoring, organized in two disordered regions, residues 1-43 and 100-148, together 92 of 183 residues; the two independent signals (sequence-based disorder prediction and structural pLDDT, mean 48.2 by metapredict/AlphaFold consensus, 40.6 by ESM-Fold) converge, the criterion we require before reporting disorder as established (Figure 2). This is a biophysical property, not a functional annotation: we draw no inference about liquid-liquid phase separation or condensate formation from disorder alone, and note that DeepTMHMM's "globular" localization call means only that the protein is not predicted to be transmembrane or secreted, not that it is a folded globular domain; with essentially half the protein disordered and both mean pLDDT scores below 50, it should not be presented as one.

Composition is where a disordered region can still be informative, provided it is compared against the right background. Since the compositional norms available for intrinsically disordered regions are derived from eukaryotic proteomes, we situated the two regions instead against the organism-matched internal reference described in Methods (2,439 predicted disordered segments of the H37Rv proteome). On the charge axes the two regions are unremarkable for *Mycobacterium tuberculosis* disorder: fraction of charged residues 0.186 and 0.163, net charge per residue 0.000 and $+0.082$, charge patterning κ 0.277 and 0.346, every value falling between the 39th and the 75th percentile of that reference. What sets them apart is a depletion of residue diversity rather than any charge signature. The combined Gly+Ala+Val fraction reaches 0.531 in the second region (94th percentile, reference median 0.302); aromatic residues are all but absent from both, none of the 43 residues of the first and two of the 49 of the second carrying an aromatic side chain; and the second region contains no proline at all, the residue that most classically promotes disorder in eukaryotic sequences, where a reference segment of that length would carry four or five, so whatever maintains disorder here, it is not that. Per-residue binomial tests against whole-proteome frequencies, corrected across all 60 tests, leave exactly two enrichments standing: serine in the first region (18.6% against 5.5%, 3.4-fold, $q = 0.042$) and glycine in the second (28.6% against 10.0%, 2.9-fold, $q = 0.004$). This refines rather than contradicts the Ala/Gly-rich description used elsewhere in this paper: taken together the Gly+Ala+Val fraction is genuinely enriched, but residue by residue that enrichment is carried by glycine alone. No tandem repeat is present, the only motif recurring three times in the whole protein being the tripeptide RRL. We report this as measured composition and draw no functional inference from it: a glycine- and serine-rich, aromatic-depleted low-diversity segment is equally compatible

with a flexible linker, with a disordered interaction segment, and with no particular role at all.

No confident homology was detected by any method applied. HHpred's best hit (DUF7993, 9.5% probability, $E = 1300$) is noise. A full local scan against Pfam-A.hmm at the curation gathering threshold returns zero hits for Rv3222c itself (as distinct from its degenerate paralogs, discussed below, in which a transposase domain is detected). Structure-based search was then pushed to the largest indices currently searchable, using both available models of the protein (the ESMFold model and AlphaFold database entry AF-O05844-F1) as queries against pdb100, CATH50, the AlphaFold/Swiss-Prot set and AFDB50, the clustered AlphaFold database of about 53 million predicted structures. The AlphaFold model recovers exactly one hit in AFDB50, itself ($E = 3 \times 10^{-27}$, probability 1.00), and nothing in pdb100, CATH50 or AlphaFold/Swiss-Prot; the self-hit is the positive control, establishing that the search was performed and was capable of returning a match. The ESMFold model, which for a protein of this kind is the less reliable of the two (mean pLDDT 40 against 48), returns two AFDB50 hits and one CATH50 hit, all at $E \geq 3$ with probabilities of 0.02 to 0.11, that is, noise. No fold relative of Rv3222c is detectable in any structural database, and the protein likewise has no match in the ESM Atlas. For a protein that is 44.8% disordered this outcome is expected rather than surprising, and we report it as a bounded negative result: it excludes a recognizable globular fold shared with any characterized structure, and it does not exclude function.

The protein is also undetected in 0 of 16 PaxDb proteomic datasets for H37Rv, reported here as an honest absence of mass-spectrometric evidence for the protein's existence, not as evidence against it. A tryptic-digest null model rules out the simplest structural explanation for that absence. Rv3222c yields 15 tryptic peptides, 5 of them in the 7-25-residue range conventionally recovered by bottom-up mass spectrometry, a low value (20.9th percentile of the H37Rv proteome) that could plausibly explain non-detection on its own. It does not: among the 244 H37Rv proteins with exactly the same count of detectable peptides, 83.2% are recovered in at least one of the same 16 PaxDb datasets, close to the proteome-wide detection rate (3,454 of 3,906 proteins, 88.4%) and far above the rate among proteins with zero or one detectable peptide (12.1% and 53.8%, respectively); a length-matched control (149-216 residues, $n = 542$) gives the same 88.4% baseline. Peptide count is therefore not the limiting factor, and Rv3222c's total absence is the exception within its own reference class rather than an expected consequence of being short and lysine/arginine-poor, a composition it does not in fact have (16 K/R residues, 8.7%, an ordinary value at the 34th percentile). The absence is better read as compatible with condition-dependent expression, since all 16 available PaxDb datasets were collected in standard growth rather than the oxidative-stress condition in which SigH, and by the co-induction reported below, plausibly Rv3222c, is active, than as a structural artefact of the protein's size or composition; we report it as an open question rather than as a third, independent pathway to invisibility alongside the absence of a detectable homolog and of a solved structure.

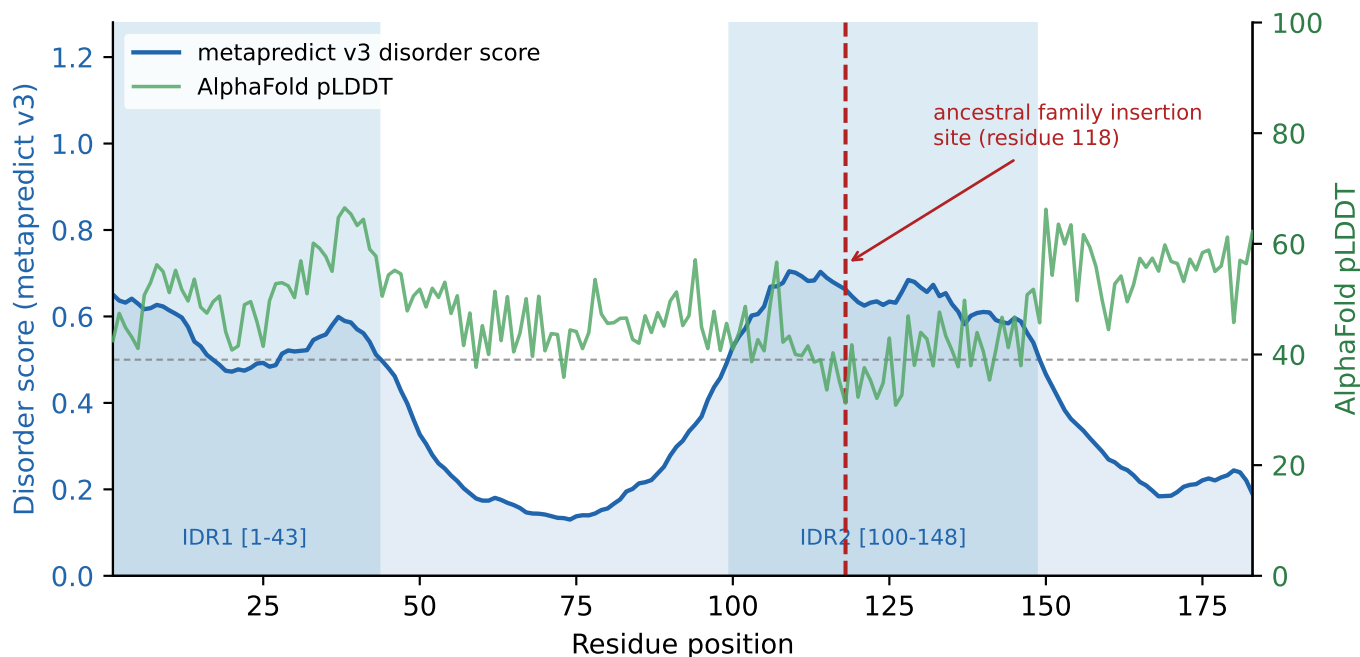
Within the genus, Rv3222c is not restricted to the MTBC, though the atlas record inherited at the start of this project stated that it was (0/53 non-MTBC genomes at a stricter, published threshold). A relaxed search (tblastn, $E < 10^{-3}$, no identity or coverage cutoff) recovers hits in 4 of 53 assembled non-MTBC *Mycobacterium* genomes, while the same search recovers *rshA* and *sigH* in 53/53 and 53/53 respectively, confirming the search itself is sensitive; the original zero reflected the stricter published identity/coverage thresholds, not an absence of detectable sequence.

A compositional null model (30 same-composition permutations of each protein) supports these as real homology rather than a compositional artefact of Rv3222c's Ala/Gly-rich sequence: 0 of 30 shuffles of either the target or the *rshA* control protein reach the corresponding real bit score, at any of the four positive species ($p = 0.032$ throughout). Two of the four hits (*M. malmoeense*, *M. heckeshornense*) fall entirely within the more compositionally biased of the protein's two halves, the N-terminal disordered region (residues 1-48, Shannon entropy 3.47 of a maximum 4.32, versus 3.82 for the C-terminal segment carrying the other two hits); because a whole-protein permutation dilutes this local bias with that of the rest of the sequence, we repeated the null model on each segment separately, shuffled at its own local composition. Both segments clear their region-specific null completely (0/30 permutations reaching the real score in either case) and are reciprocally specific to their own pair of species, a pattern a promiscuous compositional artefact, expected to cross-react similarly at both low-complexity segments across genomes that are all rich in biased-composition sequence, would not readily produce.

The strongest of the four hits is in *M. riyadhense*, the MT-BAP clade's closest known relative of the MTBC (52.4% identity, 32.2% coverage over residues 125-183, $E = 9 \times 10^{-11}$), on a large plasmid (Mr_{iy_45176_p1}, 472 kb, versus a 6.15 Mb chromosome) rather than in any operon context; its C-terminal seven residues (. . . TDTRMVHQ) match Rv3222c's exactly. This plasmid location does not itself indicate a mobilizable element: a Pfam scan of the 50 kb surrounding the hit finds no enrichment for mobility-associated domains relative to the plasmid as a whole (2 of 145 versus 30 of 1,675 predicted open reading frames), so the homolog is best read as ordinary plasmid-borne cargo rather than as evidence that the family is itself mobile.

None of the four hits sits in synteny with a *sigH/rshA* operon, which in all four species is intact and contiguous at its own locus. This absence of synteny generalizes well beyond the four species carrying a detectable homolog: across 51 of the 53 non-MTBC genomes (the remaining two, both leprosy-bacillus genomes with extensively pseudogenized sequence, gave a low-confidence *sigH* or *rshA* alignment under an a priori reliability threshold, though *M. lepromatosis* nonetheless yields a consistent result when included), *rshA* and *sigH* are directly adjacent, with a median inter-gene gap of -1 nt (range -19 to +74) and no genome exceeding 200 nt of separation, against the 544 nt occupied by Rv3222c between the same two genes in H37Rv.

What remains genuinely restricted to the MTBC, and is the narrower and better-supported claim, is the gene's position: its insertion between *rshA* and *sigH* is not observed in any tested



44.8% of residues above the 0.5 disorder threshold (mean AlphaFold pLDDT = 48.2)

Figure 2: **Per-residue disorder profile of Rv3222c.** metapredict v3 disorder score (blue, left axis) and AlphaFold pLDDT (green, right axis) across the 183 residues, shaded over the two consensus disordered regions (IDR1, IDR2). The two independent signals track each other: pLDDT dips where the disorder score peaks. The dashed red line marks residue 118, the position at which an IS110-family element inserted into a paralogous copy of this gene family (Section 3.6, not this copy), falling inside IDR2.

non-MTBC genome, even where a homologous sequence exists elsewhere. Of the recovered non-MTBC open reading frames, only one (*M. malmoense*, 344 residues) reconstructs to a length consistent with a complete protein; the others are markedly shorter than Rv3222c and one lacks an in-frame start codon upstream of its aligned region, so a length-based ortholog assignment could not be made with confidence across all four hits. Within the genus proper, reciprocal-best-hit search identifies unambiguous orthologs in *M. bovis* (Mb3249c, 100% identity) and *M. orygis* (99.5% identity).

3.6. Rv3222c is the last intact member of a gene family disrupted by an ancestral IS110-family insertion

A within-genome search for sequences related to Rv3222c in H37Rv recovers, beyond the locus itself, three degenerate copies: an intergenic fragment at 833,632-833,736 (74.1% identity over 108 nt), and two fragments inside the RefSeq-annotated, single-copy vestigial insertion-sequence element IS1607 (90.1% identity over 181 nt, and 82.5% over 354 nt); a fourth, marginal 29-nucleotide hit at 93.1% identity was not counted as a family member. None of the three degenerate copies overlaps an annotated gene. Repeating this census, restricted to complete genome assemblies to avoid inflation from assembly fragmentation, shows an identical result, four regions, three degenerate copies at 74.1%, approximately 82%, and 90.1% identity, in H37Rv, in *M. bovis* AF2122/97, and in *M. africanum* GM041182 (lineage 6) (Figure 3). These three genomes span the MTBC's

two deepest clades (the human-restricted lineages and the animal-adapted ecotypes), so an identical family content, down to the same identities to a tenth of a percent, indicates a state fixed at or before the complex's last common ancestor, not an ongoing process observed at different stages in different lineages. The *M. canettii* reference strain CIPT 140010059 (STB-A), by contrast, carries at least 14 copies (a minorant, since *M. canettii* is the most diverged from the H37Rv query used and therefore the least sensitively detected). The family is absent from *M. leprae*, and is therefore a mycobacterial, but not a genus-universal, feature.

That single outgroup genome, however, turns out not to represent the smooth tubercle bacilli as a group, and the contrast it suggests requires a more precise statement. We repeated the census on every *M. canettii* or smooth-tubercle-bacillus assembly at chromosome level or better in NCBI (29 genomes, contig N50 from 43 kb to 4.5 Mb), scoring both the number of related regions and the state of the locus itself. The locus falls into two disjoint states with no intermediate: either intact, a single 552-nt block at 99.1 to 100% identity with an *rshA*-to-*sigH* spacing of 544 nt, exactly the H37Rv configuration (10 strains), or split, the best block truncated to the gene's N-terminal 353 nt at 82.5 to 93.6% identity (19 strains, including STB-A). The spacing is strikingly uniform in the intact group: all 10 give exactly 544 nt, the H37Rv value to the nucleotide. In the split group it is dilated to between 1.1 and 1.8 kb in the seven strains whose assembly places both neighbors on the same contig with the intervening region complete, and unreliable in the remaining twelve, where

the locus is interrupted by a contig boundary and the measured spacing (308 to 593 nt) reflects assembly structure rather than genome structure. Copy number tracks the partition: 2 to 7 regions in the intact group, 5 to 16 in the split group.

Assembly quality does not account for the partition, although it does affect the count. The rank correlation between contig N50 and the number of regions is +0.29 overall, +0.13 within the intact group and +0.77 within the split group, as expected if fragmentation undercounts repeated copies without creating them. Read at matched quality the partition holds regardless: among the four assemblies with contig N50 between 700 kb and 1.03 Mb, STB-K and STB-D give 16 and 10 regions with a split locus, while STB-J and STB-L give 2 and 5 with an intact one. The decisive case is ET1291, the most complete assembly of the whole cohort, a single closed circular chromosome from hybrid long-and-short-read sequencing: it yields four regions and an intact locus, and no assembler can delete copies from a closed chromosome.

ET1291 is not an ordinary *M. canettii* strain, and this is what makes its state informative. It is the Ethiopian smooth tubercle bacillus that occupies an intermediate branch between the reconstructed common ancestor of the MTBC lineages and the closest node of the other known *M. canettii* strains, separated by 5,948 SNPs from the MTBC ancestral genome against at least 8,984 from the nearest previously described *M. canettii*, and by only 3,389 SNPs once recombination is accounted for, about twice the maximal distance between any two MTBC strains [37]. Our own ANI measurements reproduce that placement (99.53% to H37Rv against 99.31% to STB-A). The closest known evolutionary intermediate to the complex therefore already carries the MTBC state of this family: an intact gene occupying the operon, the same 544-nt spacing between its neighbors, and a handful of degenerate copies. The contraction is consequently not an event of the MTBC branch, nor a uniform ancestral property of *M. canettii*; it was already complete in the lineage leading to the complex, while the expansion to 14 or more copies characterizes a subset of the more divergent smooth bacilli, those whose recombinogenic lifestyle persists [37, 38].

The composition of that cohort has to be stated, because 29 assemblies are not 29 independent observations. Nineteen of them come from a single BioProject (PRJEB48348), which is itself enriched for the split state: 16 split against 3 intact, where the remainder of the cohort gives 3 against 7 (Fisher's exact test, $p = 0.011$). Nothing in a sequencing protocol splits a gene, so this is not a technical bias but the expected signature of phylogenetic non-independence among isolates of one collection. Collapsing near-identical strains by single-linkage clustering on ANI and retaining one representative per group, at three thresholds from 99.90 to 99.99% identity, leaves every qualitative conclusion intact: at the most aggressive collapse, 99.90%, the cohort reduces to 17 representatives, 9 intact and 8 split, the two identity ranges remain disjoint (99.1-100% against 82.5-92.5%), and the association with the dominant BioProject is no longer detectable ($p = 0.153$), confirming that it was carried by redundancy rather than by any property of the project. The rank correlation between contig N50 and region count likewise ceases to be distinguishable from zero once near-duplicates

are removed (+0.25, $p = 0.34$), which weakens the assembly-quality explanation further. The counts above are therefore reported as a census of the assemblies available, not as a sample of independent genomes.

Where the two states sit relative to one another matters more than how many strains carry each. On the neighbor-joining and UPGMA topologies built from pairwise ANI across the cohort (Methods, Figure 4), the split state is never monophyletic, in either topology or in any of 200 parametric bootstrap replicates drawn from the confidence intervals of the ANI estimates. Fitch parsimony requires a minimum of four transitions between states, a count that is invariant to rooting and identical under both topologies; that value is nonetheless far from what chance would give on this tree, since permuting the states across the tips yields a median of nine transitions and only 3 of 10,000 permutations reach four ($p = 3 \times 10^{-4}$). The character is thus strongly structured phylogenetically without being resolvable into a single clade. Under an irreversible model, which is the mechanically appropriate one here, the minimum number of independent acquisitions of the split state is four in the neighbor-joining topology (median five across bootstrap replicates): a clade of 13 Percy strains together with STB-D, a group of three including STB-A, a pair, and STB-K.

The polarity itself does not come from the tree but from the mechanism. An insertion can be acquired repeatedly, but it is not repaired: nothing restores a 552-nt reading frame at 99% identity or better, with its 544-nt spacing between neighbors, once an element of roughly one kilobase has entered it and a 21-nt deletion has accompanied the event. The intact state is therefore ancestral and the split state derived, whatever the rooting.

Four independent acquisitions are, however, difficult to reconcile with the junctions themselves. Every split strain in which the boundary is measurable shows the same cut point to the nucleotide (query positions 353 and 375), and independent insertion events would not be expected to share identical boundaries. The reading that accommodates both observations is a single ancestral insertion whose present distribution has been redrawn by homologous recombination, a process that is frequent and characteristic of the smooth bacilli, even though its rate is notoriously difficult to estimate without violating model assumptions [38, 39]. This reading is not asserted on general grounds alone: a four-gamete test on the 240 biallelic sites called across the same 32-genome alignment finds 33.9% of site pairs incompatible with a single tree, an incompatibility that grows with physical distance between sites (6.2% at 0-100 nt to 47.7% beyond 5 kb), the signature expected under recombination rather than recurrent homoplasy; a permutation-based test of compatibility clustering along the sequence confirms the pattern (observed mean neighbour incompatibility 0.275 against 0.339 under 1,000 column-order permutations, $p = 0.0010$). The test is underpowered to localize a breakpoint or to fully exclude recurrent homoplasy at any single site, a limitation expected at this sample size (32 genomes). That reading also disposes of the apparent anomaly which motivated the analysis: STB-K carries the split state while being the most divergent strain of the cohort, but its two closest relatives by ANI are Percy79 and Percy301 (98.56 and 98.50%),

both split, and its nearest patristic neighbor is a split strain in 77% of bootstrap replicates. It is not an isolated split strain among intact ones. One caveat conditions all of the above: an ANI tree is a tree of overall similarity, not a genealogy, and recombination is precisely what blurs the distinction between the two.

The family is not confined to the genus, however. An unrestricted BLASTp search against the non-redundant protein database, validated by the same search recovering both operon neighbors broadly distributed outside *Mycobacterium* (the *rshA* and *sigH* controls each return several hundred non-*Mycobacterium* hits, confirming the search itself is not blind to homology outside the genus), recovers eight non-*Mycobacterium* actinobacterial hits for Rv3222c, corresponding to seven independent genomic loci (two hits fall on the same *Micromonospora* sp. DT47 contig, entered separately under GenBank and RefSeq accessions), at moderate identity (approximately 50%, 40-104 bit score) in *Dietzia*, *Rhodococcus* and other Corynebacteriales, and in more distant Actinomycetota (*Micromonospora*, *Jatrophihabitans*); three further hits at 96.6-100% identity to distant Enterobacterales genera were excluded as almost certainly database contamination, near-perfect identity across a phylogenetic distance at which even ribosomal proteins do not reach it being far more consistent with a mislabeled or contaminated metagenome-derived record than with a genuine relationship. The family therefore predates the mycobacterial radiation and has been retained, at varying degrees of decay, across Actinobacteria broadly; its sharpest contraction, to a single functional copy, is specific to the MTBC lineage.

The internal architecture of this family resolves what the sequence-similarity census alone could not: whether Rv3222c descends from a mobile-element protein, or a mobile element instead disrupted a member of Rv3222c's own family. Two lines of evidence settle this in the second direction. First, the three fragments of the H37Rv IS1607 locus recombine, once the two intervening non-homologous segments (82 and 794 nt) are removed, into a single open reading frame of 185 residues with no internal stop codon, beginning with the same eight residues (MSSPVSSR) as Rv3222c; a chance concatenation of unrelated fragments would not be expected to produce an open, in-frame reading through 185 codons. Second, and decisively, delimiting the mobile element's boundaries by comparing its copies to one another, without reference to any existing annotation, places those boundaries within 2-4 nucleotides of the two Rv3222c-derived fragments: the gene's N-terminal 354 nt forms the element's left flank and its C-terminal 182 nt forms its right flank, in the correct order and orientation. The element (1,537 nt, positions 2,202,803-2,204,339 on the *M. canettii* reference) is therefore not merely adjacent to a degenerate gene fragment; it is bounded, on both sides, by the two halves of the same gene it interrupted (Figure 5).

The direction of this relationship, an element that inserted into a family member versus a family that arose from the element's own sequence, is settled by the position of the insertion's cut point rather than by sequence similarity, since both scenarios involve, by construction, members of the same gene family and

therefore predict similar reconstructed sequences. Across five independent copies of the element (three additional ones in *M. canettii*, plus *M. canettii*'s and H37Rv's primary copies), the left fragment ends at exactly the same nucleotide (query position 353) and the right fragment begins at exactly the same nucleotide (query position 375) in every case: zero dispersion. An independent insertion event at a preferred target site would still be expected to fall at different positions within the gene each time; identical cut points to the nucleotide, five times over, indicate a single ancestral insertion whose product then transposed as a block.

Those five copies come from one genome, however, and copies of a single event replicated by duplication are not independent observations of it. We therefore repeated the measurement across genomes rather than within one, over 31 assemblies: the 29-genome cohort above plus H37Rv and *M. africanum* lineage 6 as MTBC controls. The disrupted, non-operonic copy can be located in 29 of the 31, the two failures being fragmented assemblies in which it falls across a contig boundary. In all 29 the left fragment ends at coding-sequence position 353, and in 27 of them the right fragment resumes at position 375, the two exceptions (positions 391 and 399) being truncations of the right alignment rather than displacements of the cut. Genomes separated by tens of thousands of SNPs, on both sides of the MTBC/smooth-bacillus divergence, therefore carry the same insertion point to the nucleotide. The event predates that divergence.

The intrusion inside the operon is a separate and later matter, and it is confined to the split-locus group. None of the 12 intact-locus genomes, including both MTBC controls, shows an insertion between *rshA* and *sigH*. Among the 19 split-locus strains, the five whose assemblies keep the operonic region on a single contig all show an insert of 0.9 to 1.3 kb carrying the same 353/375 signature, that is, a copy of the same composite unit rather than a fresh cut of the gene; in the remaining fourteen the region is fragmented and the test is not applicable, so their operonic state is inferred from the truncated locus block rather than measured directly. In *M. canettii* STB-A this copy is a fourth instance of the segmental duplication described above, inserted directly between the two halves of the operonic ortholog. The operon of the MTBC, and of the smooth bacilli closest to it, was never entered by the element. A dedicated hotspot-preference control supports the same conclusion: the five copies are more similar to one another (95.5% identity) than any of them is to Rv3222c itself (90.4%), a 5.2-point gap; because this first comparison spans the whole 1,531-nt unit, of which 995 nt is transposase body shared under both candidate scenarios, we repeated it restricted to the gene-derived fragments alone, where it holds with an even larger margin (left fragment, 92.7% between copies versus 88.4% to Rv3222c; right fragment, 97.6% versus 92.3%). Insertion occurred after coding-sequence position 353 (protein residue 118, within the second disordered region), accompanied by a 21-nucleotide (7-residue, VGVGRGES) deletion not recovered elsewhere in any copy.

The element's transposase body carries a complete IS110-family catalytic domain (Pfam PF01548 DEDD_Tnp_IS110, full model coverage 2-155/155, score approximately 120, $E \approx$

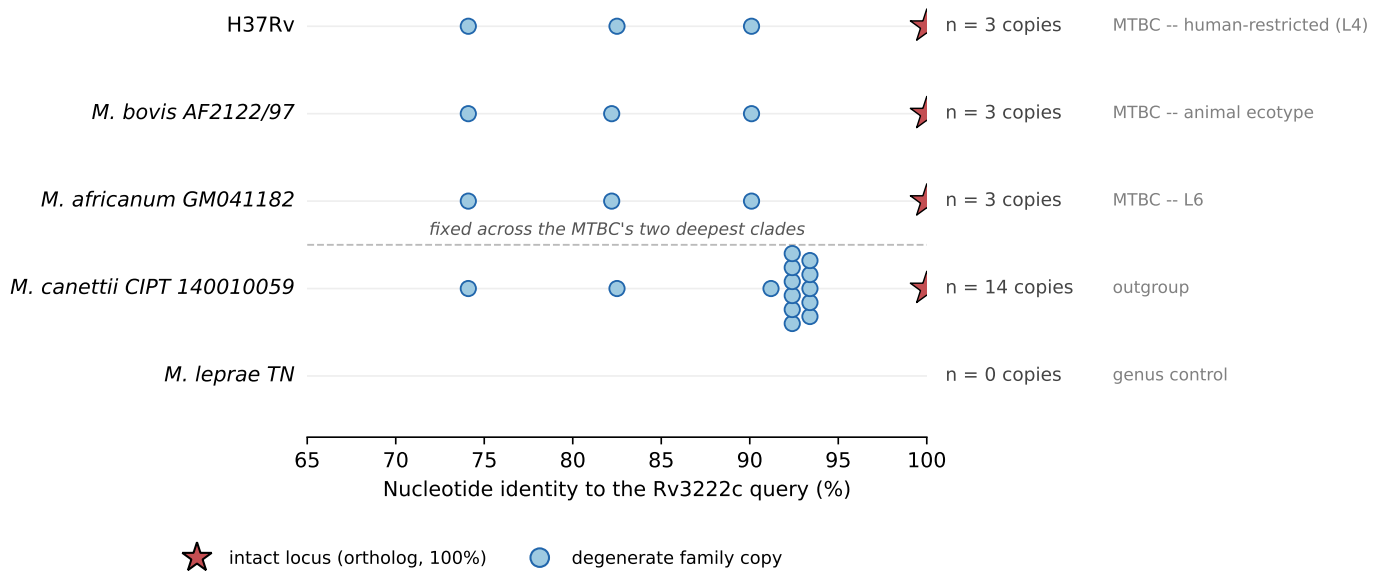


Figure 3: Family content is fixed across the MTBC and expanded in the outgroup. Each marker is one region related to Rv3222c recovered in a complete genome assembly (star: the intact locus itself, at 100% identity; circles: degenerate copies, positioned by nucleotide identity to the H37Rv query). H37Rv, *M. bovis* AF2122/97 and *M. africanum* GM041182 (lineage 6), spanning the MTBC's two deepest clades, carry an identical count and identity distribution; the *M. canettii* reference strain CIPT 140010059 (STB-A) carries 14 copies (a minorant, see main text), a state shared by some but not all smooth bacilli; *M. leprae* carries none.

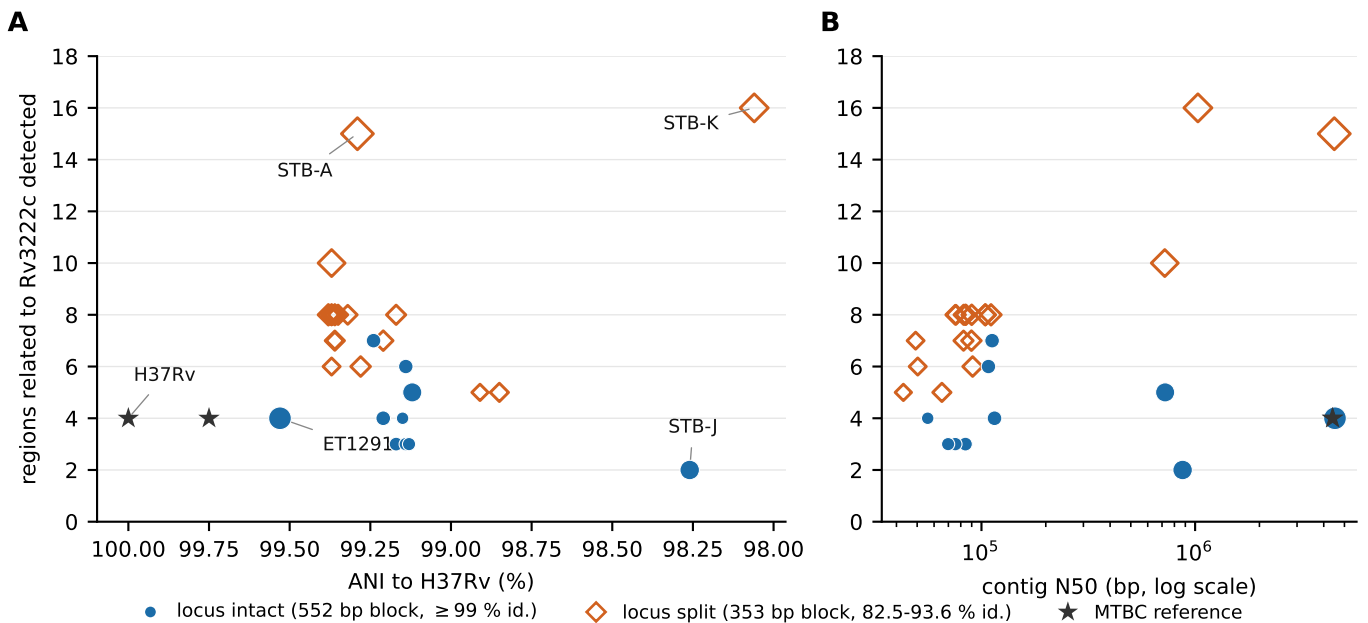


Figure 4: The state of the locus across the smooth tubercle bacilli follows neither proximity to the MTBC nor assembly quality. Each marker is one assembly of the 29-genome cohort; filled circles are strains whose locus is intact (a single 552-nt block at $\geq 99\%$ identity), open diamonds strains whose locus is split (best block truncated to the N-terminal 353 nt at 82.5-93.6%), and stars the two MTBC references, recounted here with the same metric as the cohort. Marker area scales with the logarithm of contig N50. (A) Number of regions related to Rv3222c against ANI to H37Rv: the two states interleave along the whole range of genomic proximity to the complex, and ET1291, the closest known evolutionary intermediate, carries the MTBC state. (B) The same counts against contig N50: assembly quality modulates how many copies are recovered but does not separate the two states, and the two most complete assemblies of the cohort fall on opposite sides of the partition.

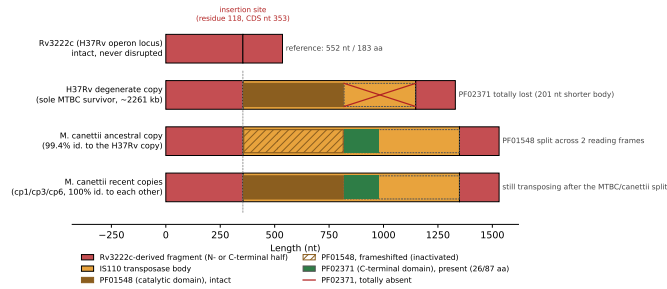


Figure 5: The composite element is bounded by the two halves of the gene it split. Top: the intact Rv3222c reference, never disrupted at its own operon locus. Below: three instances of the composite element, drawn to scale, each flanked by the same Rv3222c-derived N- and C-terminal fragments. The H37Rv copy has lost its C-terminal transposase domain (PF02371) entirely; the *M. canettii* ancestral copy (orthologous to the H37Rv copy, 99.4% identical to it) retains a truncated PF02371 but has its catalytic domain (PF01548) split across two reading frames; the more recent *M. canettii* copies (cp1/cp3/cp6, identical to one another) retain an intact PF01548 and the same truncated PF02371, consistent with continued transposition after the MTBC/*M. canettii* split.

5×10^{-38} in both H37Rv and *M. canettii*), the domain family that defines bridge-RNA-guided, site-specific DNA recombinases [40, 41]. Its associated C-terminal domain (Pfam PF02371) is present, but truncated to 26 of 87 residues, in *M. canettii*, and entirely absent in H37Rv, a difference that tracks a 201-nucleotide length difference in the transposase body between the two lineages (995 versus 794 nt) and a corresponding increase in an internal, frameshifted, and separately inactivated copy. Fine-grained comparison of individual *M. canettii* copies shows two distinct fates layered on top of the single ancestral event: one copy (99.4% identical to the H37Rv copy despite genome-wide MTBC/*M. canettii* divergence of 0.2-0.3% at housekeeping genes) is the shared ancestral copy, inherited unchanged in position since before the split and independently inactivated by a reading-frame shift within its catalytic domain; three further copies are identical to one another at 100% and to the ancestral group at 87.9%, indicating transposition events specific to the *M. canettii* lineage after the split. The element is thus active, and was recently active, in *M. canettii*; in the MTBC, where the family is otherwise fixed and unchanged across the deepest available clade split, it is represented by a single, doubly degraded copy that has not moved since.

This element corresponds to IS1607, described in 2000 from a single copy shared by *Mycobacterium tuberculosis*, *M. bovis*, BCG, *M. africanum* and *M. microti*, and reported absent from 69 strains across 19 non-tuberculous mycobacterial species [42]; on that basis the original description concluded the element was specific to the MTBC. Its family assignment here rests on Pfam, not on ISfinder: IS1607 is absent from the complete 2020 snapshot of the ISfinder sequence set, and no archived record page for it exists. The original description is itself consistent with this absence, since it presents IS1607 not as an insertion sequence but as an “insertion sequence-related element”, already degenerate and present in a single copy at the time of its description, and such an element would not have been submitted for nomenclature. *M. canettii* was not available as a genome sequence at the

time and was accordingly absent from that panel. Our results indicate the opposite history: IS1607 is an ancestral mycobacterial element, still active in the outgroup, that went extinct within the MTBC, leaving behind the single degraded copy that disrupted a member of the Rv3222c family before the complex’s internal diversification, itself confirmed by presence of the same insertion, at the same 794-nucleotide flank distance, in both H37Rv and *M. bovis* AF2122/97, and, in *M. canettii*, by an 881-nucleotide flank distance consistent with indel accumulation since the split. This correction does not diminish the original finding; Dziadek et al.’s [42] conclusion was reasonable given the genomes available in 2000, and is superseded here specifically by the subsequent availability of the *M. canettii* genome, an evidentiary update rather than a methodological error on their part.

Finally, in *M. canettii*, the direct ortholog of Rv3222c itself (as distinct from the degenerate family members discussed above) occupies the same position between *rshA* and *sigH*, with the same short overlaps; its C-terminal half is identical to H37Rv’s over 178 nt, a higher identity than that of the neighboring, well-conserved *sigH* (97.7%) despite an overall genome-wide divergence of only 0.2-0.3%. The *rshA-sigH* intergenic span itself is 1,774 nt in *M. canettii* against 544 nt in H37Rv, consistent with the additional family-related sequence described above having accumulated in this region in the outgroup.

4. Discussion

4.1. A first handle on function, independent of genomic context

Every functional argument advanced so far, co-transcription inferred from operon structure, the STRING neighborhood and coexpression scores, and regulation by SigH itself, ultimately encodes the same single fact: genomic adjacency. An independent transcriptomic dataset offers a first handle that does not. Under cigarette-smoke-condensate exposure, an oxidative-stress-inducing condition, Rv3222c is induced at log2 fold-change 1.03 ($p = 2.4 \times 10^{-9}$) in the same experiment in which *sigH* itself is induced at 1.04 ($p = 3.7 \times 10^{-9}$) [33]. The near-identical magnitude and significance of the two responses is consistent with co-regulation under the same oxidative-stress signal that activates SigH, and is, to our knowledge, the first piece of evidence for Rv3222c’s behavior under stress that does not derive from its position in the genome. It remains, however, a correlative, single-condition observation: it establishes co-expression, not a physical or functional interaction, and should be weighted accordingly against the essentiality and evolutionary evidence above, not substituted for it.

4.2. An experimentally tractable target already in hand

The depletion strain introduced in Section 3.2 was screened across six chemical-genetic screens and retained in the platform’s higher-quality strain subset [29]. Restricted to the 91 strains sharing the same promoter strength, its baseline growth (4.09 population doublings) sits close to the panel median (4.05, 53rd percentile), comparable to *rpoB* at the same promoter (4.14). We looked for, but do not report as a finding, a specific chemical sensitization signature: the strain’s response to its best-matching

compound is unremarkable against the full 340-strain panel (rank 45-155 of 340 depending on assay, z never below -1.02), and although it stands out in ten of 140 reference conditions (concentrated on QcrB, RpoB and GyrAB inhibitors), all 140 of those reference conditions are themselves strongly active compounds (median scaled growth ratio below -2), so the public dataset offers no weakly active contrast against which to distinguish a pathway-specific sensitization from general fragility under strong chemical stress. Only the platform’s full primary screening data, held by its originating group, could resolve this question; what can be reported with confidence is narrower but still useful: a chemogenomically characterized, tractable depletion strain for this gene already exists, independently confirming its essentiality and offering a direct route to further experimental work without requiring new strain construction.

A second set of strains, built for entirely different purposes, offers an even cheaper first experiment. The $\Delta sigH$ mutants of Section 3.3 all carry an intact Rv3222c downstream of an interrupted *sigH*, and, on the transcriptional architecture reported here, no promoter of their own to express it from. Whether the gene is nonetheless transcribed in those strains is a single RT-qPCR away, on material that already exists in at least three laboratories, and the answer is informative either way: transcription would establish read-through across the resistance cassette and close the apparent paradox; its absence would place the essentiality of Rv3222c itself in question in those backgrounds, which is exactly the kind of test a claim of essentiality should be exposed to. The same strains would also allow the converse experiment, complementing with *sigH* alone rather than with the operon fragment both original studies happened to use.

4.3. Two literature errors, correctly weighted

This study corrects two errors in the published record concerning this locus, and they warrant different treatment. The first, Thakur et al.’s [32] substitution of Rv3222c for Rv3221A/RshA in a table of sigma/anti-sigma pairs, is a transcription error with no biological consequence for that paper’s own results, which concern RshA’s actual promoter-recognition domains; we report it because it is the sole source of every subsequent literature mention of Rv3222c, including an inherited annotation of “sigma factor associated” in Willemse et al. [33] and a matching data identifier in Bond et al. [29], and because its narrow propagation pattern, present in computationally manipulated locus-tag tables and absent from prose review literature, is itself informative about how an unnamed gene is uniquely exposed to this class of error. The second, Dziadek et al.’s [42] conclusion that IS1607 is specific to the MTBC, is of an entirely different kind: not a transcription error but an evidentiary update (Section 3.6), which we correct without attributing any fault to the original analysis.

4.4. Scope and limits of the claims

This study is an *in silico* characterization and literature re-derivation, not an experimental one, and its claims should be read at that resolution. Essentiality is well supported, by two independent transposon-sequencing analyses and by the existence of a third group’s depletion strain. Its apparent non-uniformity

across MTBC lineages (Section 3.2), however, is not a clean finding: the large-scale duplication described there tracks the significant/non-significant split almost exactly. The values used here come from an independent reprocessing of the raw sequencing data rather than the authors’ own output, so they are not simply wrong, but a copy-number artefact is a plausible full or partial explanation that the present data cannot exclude. This pattern should therefore be read as an intriguing, unresolved observation, not as an established lineage-dependent essentiality difference, until it is tested against a duplication-aware reanalysis or independent Tn-seq data. Extreme CRISPR-interference vulnerability is not supported, and the claim is retracted here rather than carried forward; nothing above should be read as implying it. Intrinsic disorder is a measured biophysical property, not a functional annotation, and licenses no claim about condensate formation or liquid-liquid phase separation. No physical interaction with SigH or RshA has been demonstrated: every STRING partner score for Rv3222c lacks experimental or curated-database support, a direct IntAct query returns no interaction record for the protein at all, and co-induction with *sigH* establishes correlation, not mechanism. Consequently, the strongest claim this evidence supports is that Rv3222c is a genuine candidate for functional characterization within the SigH-RshA stress response, essential, evolutionarily constrained, and now experimentally tractable via an existing depletion strain; it does not, on the present evidence, support a drug-target claim, which would require at minimum a demonstrated druggable pocket, something the protein’s substantial disorder makes an *a priori* difficult proposition rather than a foregone one.

5. Conclusion

Rv3222c has been essential, highly conserved, and physically embedded in one of *Mycobacterium tuberculosis*’s best-studied regulatory loci for as long as its genome has been sequenced, and has never itself been the subject of a single publication, a fact traceable in part to a single locus-tag error that has substituted for actual study in every reference the gene has received before this one. Systematically re-deriving the claims made about it shows that some of the strongest, most cited motivations for studying it as a drug target, extreme CRISPR-interference vulnerability and restriction to the MTBC, do not survive examination of their underlying data, while its essentiality, its measured intrinsic disorder, and above all its evolutionary history do. That history is now established end to end: a small gene family present since before the divergence of the MTBC from *M. canettii*, invaded once, in an ancestor common to both lineages, by an IS110-family recombinase that split one member’s coding sequence in two, still active and still transposing in the outgroup, extinct and fixed at a single degraded copy across the entire MTBC. Rv3222c is the only member of this family that this history left untouched, essential where its few degenerate relatives are silent, and it is, on the evidence assembled here, a well-motivated and now experimentally tractable candidate for the functional characterization it has never received.

CRedit authorship contribution statement

Christophe Guyeux: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used Claude Code (Anthropic), an AI-agent software environment powered by the Claude Sonnet 5 model, for in silico data retrieval, statistical analysis, literature verification and drafting assistance, as described in the Materials and Methods. After using this tool, the author reviewed, independently verified against primary sources, and edited the content as necessary, and takes full responsibility for the content of this publication.

Data availability

All data analyzed in this study are publicly available and cited throughout the Materials and Methods (accession numbers, DOIs and supplementary-data sources are given at each use). Analysis scripts are available from the corresponding author upon reasonable request and will be deposited in a public repository upon acceptance.

References

- [1] T. Song, S. L. Dove, K. H. Lee, R. N. Husson, RshA, an anti-sigma factor that regulates the activity of the mycobacterial stress response sigma factor SigH, *Molecular Microbiology* 50 (3) (2003) 949–959. doi:10.1046/j.1365-2958.2003.03739.x.
- [2] J. D. Sharp, A. K. Singh, S. T. Park, A. Lyubetskaya, M. W. Peterson, A. L. C. Gomes, L.-P. Potluri, S. Raman, J. E. Galagan, R. N. Husson, Comprehensive definition of the SigH regulon of *Mycobacterium tuberculosis* reveals transcriptional control of diverse stress responses, *PLoS ONE* 11 (3) (2016) e0152145. doi:10.1371/journal.pone.0152145.
- [3] S. T. Park, C.-M. Kang, R. N. Husson, Regulation of the SigH stress response regulon by an essential protein kinase in *Mycobacterium tuberculosis*, *Proceedings of the National Academy of Sciences* 105 (35) (2008) 13105–13110, pMID 18728196; verified 2026-08-10 via CrossRef and PubMed. doi:10.1073/pnas.0801143105.
- [4] L. Cioetto-Mazzabò, F. Boldrin, C. Beauvineau, M. Speth, A. Marina, A. Namouchi, G. Segafreddo, M. Cimino, S. Favre-Rochex, S. Balasingham, B. Trastoy, H. Munier-Lehmann, G. Griffiths, B. Gicquel, M. E. Guerin, R. Manganello, N. Alonso-Rodriguez, SigH stress response mediates killing of *Mycobacterium tuberculosis* by activating nitronaphthofuran prodrugs via induction of Mrx2 expression, *Nucleic Acids Research* 51 (1) (2023) 144–165, pMID 36546765; verified 2026-08-10 via PubMed and Europe PMC. doi:10.1093/nar/gkac1173.
- [5] M. S. Alam, M. S. Khatun, B. Yusuf, L. Li, B. Aweke Mulu, H. A. Tadesse, J. Zhang, X. Tian, C. Fang, Y. Gao, Z. Liu, J. Hu, X. Chen, N. Zhong, L. Feng, H. M. A. Hameed, S. Wang, T. Zhang, The stress response factor SigH mediates intrinsic resistance to multiple antibiotics in *Mycobacterium abscessus*, *Microbiology Spectrum* 14 (7) (2026) e0396525, pMID 42294711; verified 2026-08-10 via PubMed and Europe PMC (PMC13340314). doi:10.1128/spectrum.03965-25.
- [6] G. Arora, C. W. Munson, M. Ahmed, V. Shivanna, A. Devi, V. S. Devireddy, B. Antony, S. Hall-Ursone, O. D. Gonzalez, E. J. Dick, C. Jagannath, X. Alvarez, S. Mehra, S. A. Khader, D. K. Singh, D. Kaushal, Development and pre-clinical evaluation of next-generation Δ sigh-based live candidate vaccines, *JCI Insight* 10 (19) (2025) e195947, pMID 40875531; verified 2026-08-10 via PubMed and Europe PMC (PMC12513488). doi:10.1172/jci.insight.195947.
- [7] M. A. DeJesus, E. R. Gerrick, W. Xu, S. W. Park, J. E. Long, C. C. Boutte, E. J. Rubin, D. Schnappinger, S. Ehrh, S. M. Fortune, C. M. Sasseti, T. R. Ioerger, Comprehensive essentiality analysis of the *Mycobacterium tuberculosis* genome via saturating transposon mutagenesis, *mBio* 8 (1) (2017) e02133–16. doi:10.1128/mBio.02133-16.
- [8] B. Bosch, M. A. DeJesus, N. C. Poulton, W. Zhang, C. A. Engelhart, A. Zaveri, S. Lavalette, N. Ruecker, C. Trujillo, J. B. Wallach, S. Li, S. Ehrh, B. T. Chait, D. Schnappinger, J. M. Rock, Genome-wide gene expression tuning reveals diverse vulnerabilities of *M. tuberculosis*, *Cell* 184 (17) (2021) 4579–4592.e24. doi:10.1016/j.cell.2021.06.033.
- [9] G. Senelle, M. R. Sahal, K. La, T. Billard-Pomares, J. Marin, F. Mougari, A. Bridier-Nahmias, E. Carbonnelle, E. Cambau, G. Refrégier, C. Guyeux, C. Sola, Towards the reconstruction of a global TB history using a new pipeline "TB-Annotator", *Tuberculosis* 143 (2023) 102376. doi:10.1016/j.tube.2023.102376.
- [10] S. T. Cole, R. Brosch, J. Parkhill, T. Garnier, C. Churcher, D. Harris, et al., Deciphering the biology of *Mycobacterium tuberculosis* from the complete genome sequence, *Nature* 393 (1998) 537–544. doi:10.1038/31159.
- [11] A. Kapopoulou, J. M. Lew, S. T. Cole, The MycoBrowser portal: a comprehensive and manually annotated resource

- for mycobacterial genomes, *Tuberculosis* 91 (1) (2011) 8–13. doi:10.1016/j.tube.2010.09.006.
- [12] C. Camacho, G. Coulouris, V. Avagyan, N. Ma, J. Papadopoulos, K. Bealer, T. L. Madden, BLAST+: architecture and applications, *BMC Bioinformatics* 10 (2009) 421. doi:10.1186/1471-2105-10-421.
- [13] A. Jinich, A. Zaveri, M. A. DeJesus, A. Spencer, R. Almada-Monter, E. Flores-Bautista, C. M. Smith, C. M. Sasseti, J. M. Rock, S. Ehrh, D. Schnappinger, T. R. Ioerger, K. Y. Rhee, The Mycobacterium tuberculosis transposon sequencing database (MtbTnDB): A large-scale guide to genetic conditional essentiality, *Molecular Microbiology* 124 (1) (2025) 91–101. doi:10.1111/mmi.15370.
- [14] A. F. Carey, J. M. Rock, I. V. Krieger, M. R. Chase, M. Fernandez-Suarez, S. Gagneux, J. C. Sacchettini, T. R. Ioerger, S. M. Fortune, TnSeq of Mycobacterium tuberculosis clinical isolates reveals strain-specific antibiotic liabilities, *PLoS Pathogens* 14 (3) (2018) e1006939. doi:10.1371/journal.ppat.1006939.
- [15] B. Weiner, J. Gomez, T. C. Victor, R. M. Warren, A. Sloutsky, B. B. Plikaytis, J. E. Posey, P. D. van Helden, N. C. Gey van Pittius, M. Koehrsen, P. Sisk, C. Stolte, J. White, S. Gagneux, B. Birren, D. Hung, M. Murray, J. Galagan, Independent large scale duplications in multiple M. tuberculosis lineages overlapping the same genomic region, *PLoS ONE* 7 (2) (2012) e26038. doi:10.1371/journal.pone.0026038.
- [16] J. M. Lotthammer, J. Hernández-García, D. Griffith, D. Weijers, A. S. Holehouse, R. J. Emenecker, Metapredict enables accurate disorder prediction across the tree of life, *bioRxiv* (2024). doi:10.1101/2024.11.05.622168.
- [17] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, et al., Highly accurate protein structure prediction with AlphaFold, *Nature* 596 (2021) 583–589. doi:10.1038/s41586-021-03819-2.
- [18] L. Zimmermann, A. Stephens, S.-Z. Nam, D. Rau, J. Kübler, M. Lozajic, F. Gabler, J. Söding, A. N. Lupas, V. Alva, A completely reimplemented MPI bioinformatics toolkit with a new HHpred server at its core, *Journal of Molecular Biology* 430 (15) (2018) 2237–2243. doi:10.1016/j.jmb.2017.12.007.
- [19] M. van Kempen, S. S. Kim, C. Tumescheit, M. Mirdita, J. Lee, C. L. M. Gilchrist, J. Söding, M. Steinegger, Fast and accurate protein structure search with Foldseek, *Nature Biotechnology* 42 (2023) 243–246. doi:10.1038/s41587-023-01773-0.
- [20] J. Hallgren, K. D. Tsirigos, M. D. Pedersen, J. J. A. Armenteros, P. Marcatili, H. Nielsen, A. Krogh, O. Winther, DeepTMHMM predicts alpha and beta transmembrane proteins using deep neural networks, *bioRxiv* (2022). doi:10.1101/2022.04.08.487609.
- [21] R. K. Das, R. V. Pappu, Conformations of intrinsically disordered proteins are influenced by linear sequence distributions of oppositely charged residues, *Proceedings of the National Academy of Sciences* 110 (33) (2013) 13392–13397, pMID 23901099; verified 2026-08-10 via CrossRef. doi:10.1073/pnas.1304749110.
- [22] T. Cortes, O. T. Schubert, G. Rose, K. B. Arnvig, I. Comas, R. Aebersold, D. B. Young, Genome-wide mapping of transcriptional start sites defines an extensive leaderless transcriptome in Mycobacterium tuberculosis, *Cell Reports* 5 (4) (2013) 1121–1131, pMID 24268774; verified 2026-08-10 via PubMed and supplementary tables (PMC3898074). doi:10.1016/j.celrep.2013.10.031.
- [23] S. S. Shell, J. Wang, P. Lapierre, M. Mir, M. R. Chase, M. M. Pyle, R. Gawande, R. Ahmad, D. A. Sarracino, T. R. Ioerger, S. M. Fortune, K. M. Derbyshire, J. T. Wade, T. A. Gray, Leaderless transcripts and small proteins are common features of the mycobacterial translational landscape, *PLoS Genetics* 11 (11) (2015) e1005641, pMID 26536359; verified 2026-08-10 via PubMed and supplementary tables (PMC4633059). doi:10.1371/journal.pgen.1005641.
- [24] D. Kaushal, B. G. Schroeder, S. Tyagi, T. Yoshimatsu, C. Scott, C. Ko, L. Carpenter, J. Mehrotra, Y. C. Manabe, R. D. Fleischmann, W. R. Bishai, Reduced immunopathology and mortality despite tissue persistence in a Mycobacterium tuberculosis mutant lacking alternative sigma factor, SigH, *Proceedings of the National Academy of Sciences* 99 (12) (2002) 8330–8335, pMID 12060776; verified 2026-08-10 via Europe PMC and full text. doi:10.1073/pnas.102055799.
- [25] A. D’Halluin, P. Polgar, T. Kipkorir, Z. Patel, T. Cortes, K. B. Arnvig, Premature termination of transcription is shaped by Rho and translated uORFs in Mycobacterium tuberculosis, *iScience* 26 (4) (2023) 106465, pMID 37096044; verified 2026-08-11 via PubMed. doi:10.1016/j.isci.2023.106465.
- [26] K. J. Minch, T. R. Rustad, E. J. R. Peterson, J. Winkler, D. J. Reiss, S. Ma, M. Hickey, W. Brabant, B. Morrison, S. Turkarslan, C. Mawhinney, J. E. Galagan, N. D. Price, N. S. Baliga, D. R. Sherman, The DNA-binding network of Mycobacterium tuberculosis, *Nature Communications* 6 (2015) 5829. doi:10.1038/ncomms6829.
- [27] T. R. Rustad, K. J. Minch, S. Ma, J. K. Winkler, S. Hobbs, M. Hickey, W. Brabant, S. Turkarslan, N. D. Price, N. S. Baliga, D. R. Sherman, Mapping and manipulating the Mycobacterium tuberculosis transcriptome using a transcription factor overexpression-derived regulatory network, *Genome Biology* 15 (2014) 502. doi:10.1186/s13059-014-0502-3.
- [28] H. Salgado, G. Moreno-Hagelsieb, T. F. Smith, J. Collado-Vides, Operons in Escherichia coli: genomic analyses

- and predictions, *Proceedings of the National Academy of Sciences* 97 (2000) 6652–6657. doi:10.1073/pnas.110147297.
- [29] A. N. Bond, M. Orzechowski, S. Zhang, I. Ben-Zion, A. Lemmer, N. Garry, K. Lee, M. Chen, K. Delano, E. Gath, A. L. Golas, R. Nietupski, M. Fitzgerald, S. Ehrt, E. J. Rubin, C. M. Sassetti, D. Schnappinger, N. Shores, D. K. Hunt, J. E. Gomez, D. T. Hung, Reference-based chemical-genetic interaction profiling to elucidate small molecule mechanism of action in *Mycobacterium tuberculosis*, *Nature Communications* 16 (2025) 9673. doi:10.1038/s41467-025-64662-x.
- [30] N. D. Fernandes, Q.-L. Wu, D. Kong, X. Puyang, S. Garg, R. N. Husson, A mycobacterial extracytoplasmic sigma factor involved in survival following heat shock and oxidative stress, *Journal of Bacteriology* 181 (14) (1999) 4266–4274, pMID 10400584; verified 2026-08-10 via Europe PMC and full text PMC93928. doi:10.1128/jb.181.14.4266-4274.1999.
- [31] T. Busche, R. Šilar, M. Pičmanová, M. Pátek, J. Kalinowski, Transcriptional regulation of the operon encoding stress-responsive ECF sigma factor SigH and its anti-sigma factor RshA, and control of its regulatory network in *Corynebacterium glutamicum*, *BMC Genomics* 13 (2012) 445, pMID 22943411; verified 2026-08-10 via Europe PMC. doi:10.1186/1471-2164-13-445.
- [32] K. G. Thakur, A. M. Joshi, B. Gopal, Structural and biophysical studies on two promoter recognition domains of the extra-cytoplasmic function sigma factor σ^C from *Mycobacterium tuberculosis*, *Journal of Biological Chemistry* 282 (7) (2007) 4711–4718. doi:10.1074/jbc.M606283200.
- [33] D. Willemse, C. Moodley, S. Mehra, D. Kaushal, Transcriptional response of *Mycobacterium tuberculosis* to cigarette smoke condensate, *Frontiers in Microbiology* 12 (2021) 744800. doi:10.3389/fmicb.2021.744800.
- [34] R. Manganelli, M. I. Voskuil, G. K. Schoolnik, E. Dubnau, M. Gomez, I. Smith, Role of the extracytoplasmic-function sigma factor sigma^h in *Mycobacterium tuberculosis* global gene expression, *Molecular Microbiology* 45 (2) (2002) 365–374, pMID 12123450; verified 2026-08-10 via PubMed. doi:10.1046/j.1365-2958.2002.03005.x.
- [35] E. J. Rubin, B. J. Akerley, V. N. Novik, D. J. Lampe, R. N. Husson, J. J. Mekalanos, In vivo transposition of mariner-based elements in enteric bacteria and mycobacteria, *Proceedings of the National Academy of Sciences* 96 (4) (1999) 1645–1650, pMID 9990078; PMID PMC15546; verified 2026-08-11 via PubMed. doi:10.1073/pnas.96.4.1645.
- [36] X. Ju, S. Li, R. Froom, L. Wang, M. Lilic, M. Delbeau, E. A. Campbell, J. M. Rock, S. Liu, Incomplete transcripts dominate the *Mycobacterium tuberculosis* transcriptome, *Nature* 627 (8003) (2024) 424–430, pMID 38418874; GEO GSE211992; verified 2026-08-11 via CrossRef. doi:10.1038/s41586-024-07105-9.
- [37] B. Yenew, A. Ghodousi, G. Diriba, E. Tesfaye, A. M. Cabibbe, M. Amare, S. Moga, A. Alemu, B. Dagne, W. Sinshaw, H. Mollalign, A. Meaza, M. Tadesse, D. F. Gamtesa, Y. Abebaw, G. Seid, B. Zerihun, M. Getu, M. Chiacchiarretta, C. Gaudin, M. Marceau, X. Didelot, G. Tolera, S. Abdella, A. Kebede, M. Getahun, Z. Mehammed, P. Supply, D. M. Cirillo, A smooth tubercle bacillus from Ethiopia phylogenetically close to the *Mycobacterium tuberculosis* complex, *Nature Communications* 14 (1) (2023) 7519, pMID 37980337; verified 2026-08-10 via Europe PMC and full text PMC10657438. doi:10.1038/s41467-023-42755-9.
- [38] P. Supply, M. Marceau, S. Mangenot, D. Roche, C. Rouanet, V. Khanna, L. Majlessi, A. Criscuolo, J. Tap, A. Pawlik, L. Fiette, M. Orgeur, M. Fabre, C. Parmentier, W. Frigui, R. Simeone, E. C. Boritsch, A.-S. Debrerie, E. Willery, D. Walker, M. A. Quail, L. Ma, C. Bouchier, G. Salvignol, F. Sayes, A. Cascioferro, T. Seemann, V. Barbe, C. Locht, M.-C. Gutierrez, C. Leclerc, S. D. Bentley, T. P. Stinear, S. Brisse, C. Médigue, J. Parkhill, S. Cruveiller, R. Brosch, Genomic analysis of smooth tubercle bacilli provides insights into ancestry and pathoadaptation of *Mycobacterium tuberculosis*, *Nature Genetics* 45 (2) (2013) 172–179, pMID 23291586; verified 2026-08-10 via Europe PMC and CrossRef. doi:10.1038/ng.2517.
- [39] S. Sabin, A. Y. Morales-Arce, S. P. Pfeifer, J. D. Jensen, The impact of frequently neglected model violations on bacterial recombination rate estimation: a case study in *Mycobacterium canettii* and *Mycobacterium tuberculosis*, *G3 Genes|Genomes|Genetics* 12 (5) (2022) jkac055, pMID 35253851; verified 2026-08-10 via PubMed and Europe PMC (PMC9073693). doi:10.1093/g3journal/jkac055.
- [40] M. G. Durrant, N. T. Perry, J. J. Pai, A. R. Jangid, J. S. Athukoralage, M. Hiraizumi, et al., Bridge RNAs direct programmable recombination of target and donor DNA, *Nature* (2024). doi:10.1038/s41586-024-07552-4.
- [41] M. Hiraizumi, N. T. Perry, M. G. Durrant, T. Soma, N. Nagahata, S. Okazaki, et al., Structural mechanism of bridge RNA-guided recombination, *Nature* (2024). doi:10.1038/s41586-024-07570-2.
- [42] J. Dziadek, I. Wolińska, A. Sajduda, A. Dela, J. McFadden, IS1607, a single-copy insertion sequence-related element of the *Mycobacterium tuberculosis* complex, *The International Journal of Tuberculosis and Lung Disease* 4 (11) (2000) 1078–1081.